Reassessing Qualitative Self-Assessments and Experimental Validation*

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Abstract

Qualitative self-assessments of economic preferences have recently gained popularity, often supported by experimental validation, a method that links them to choices in incentivized elicitations. We illustrate theoretically that experimental validation may fail to produce reliable new measures. Empirically, analyzing data from over 13,000 participants across diverse samples, we document four key findings. First, qualitative self-assessments and traditional incentivized measures exhibit weak correlations, even when accounting for response noise. Second, qualitative self-assessments sometimes correlate more strongly with theoretically distinct incentivized elicitations than those for which they are intended to proxy. Third, relationships between qualitative self-assessments and various attributes—including geographical location, demographics, and behaviors—are unrelated to variation in incentivized elicitations. Fourth, lower cognitive ability is associated with greater reliance on response heuristics, raising additional concerns about inference from qualitative self-assessments.

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1 Introduction

Measuring preferences using choices in incentivized elicitations has been a cornerstone of experimental economics. Over the past 15 years, however, there has been growing interest in using qualitative self-assessments of preferences as a replacement for the choice-based approach. This shift has been justified by experimental validations, which demonstrate that self-assessments predict choices in incentivized elicitations. For example, participants' certainty equivalents for a risky lottery are correlated with their responses to the question, "How willing are you to take risks, in general?" (Falk et al., 2023).[1] The impact of experimental validation and qualitative self-assessments on the economics literature has been substantial: collectively, the three primary papers on the topic (Dohmen et al., 2011; Falk et al., 2018, 2023) have been cited over 9,000 times.[2] Nonetheless, it remains to be established whether experimental validation effectively identifies useful alternative measures. In particular, it is not yet clear whether experimentally-validated qualitative self-assessments yield findings comparable to those derived from incentivized elicitations.

In this paper, we examine the experimental validation method and its implications for qualitative self-assessments. We demonstrate theoretically that experimental validation may fall short of producing reliable substitute measures. Empirically, we evaluate qualitative self-assessments for six preference domains—risk, impatience, altruism, trust, reciprocity, and willingness to punish unfair behavior—by bringing together eight datasets, covering two representative samples of the U.S., three student samples, a large U.K. low-education sample,

[1] While the term "experimental validation" has roots in psychology, Falk et al. (2023) appear to have popularized it in economics, providing what we believe to be the clearest and most developed implementation of the procedure. Researchers have built on this work to develop experimentally-validated measures for ambiguity attitudes (Cavatorta and Schröder, 2019), moral universalism (Enke et al., 2022), debt aversion (Albrecht and Meissner, 2022), competitiveness (Fallucchi et al., 2020; Buser et al., 2024), preferences for truth telling (Schudy et al., 2025), and social preferences (Epper and Mitrouchev, 2025), among others.

[2] Dohmen et al. (2011) use a qualitative self-assessment to measure risk attitudes. Falk et al. (2018) and Falk et al. (2023) develop and implement preference indices that combine qualitative self-assessments with questions involving hypothetical incentives or scenarios. We focus on the qualitative self-assessments and the role of experimental validation, although the concerns we raise also apply to indices that include qualitative self-assessments, as discussed in Section 2. The practical relevance of these issues is substantial. Together with several research assistants, we examined 3,000 papers citing these three studies and found that 49.4% use data from at least one qualitative self-assessment.

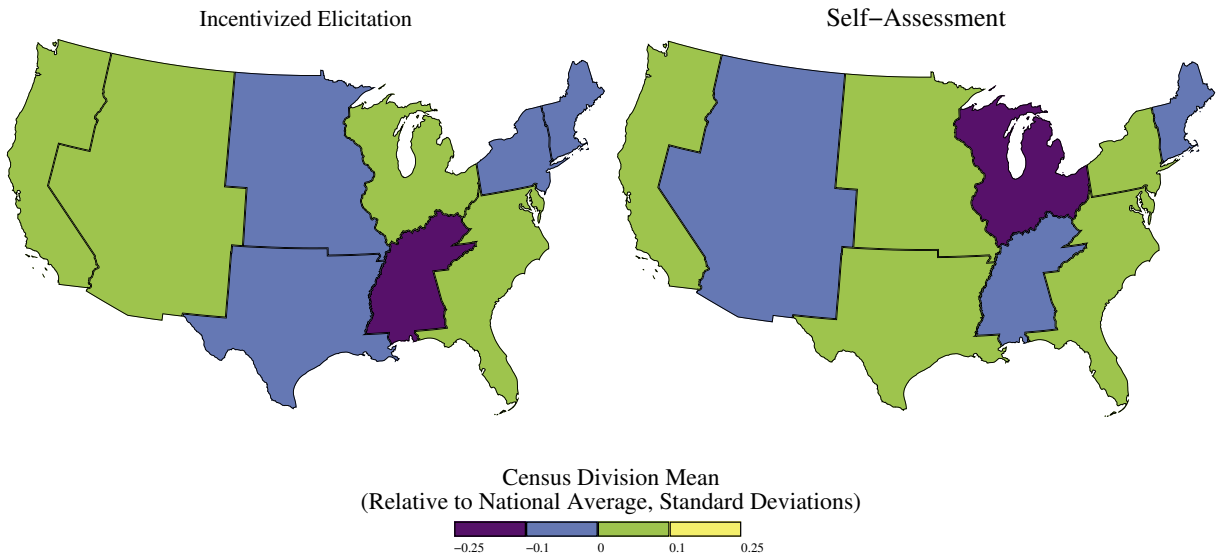
and two convenience samples (total $N \approx 13{,}000$). Each dataset contains both incentivized preference elicitations and experimentally-validated self-assessments drawn from the survey modules developed by Falk et al. (2023). Using these data, we find smaller correlations between qualitative self-assessments and their incentivized counterparts than Falk et al.'s original experimental validation, in line with prior replications in student populations. We also illustrate that qualitative self-assessments are often correlated with multiple incentivized elicitations, making it difficult to determine which preferences, if any, these self-assessments are measuring. Indeed, we show that correlations between qualitative self-assessments and economic, demographic, and geographic variables are unrelated to variation in incentivized elicitations. Finally, use of response heuristics when answering qualitative self-assessments is correlated with cognitive ability. The link between heuristics and demographics may further confound estimated relationships.

The implications of these findings are evident from Figure 1. The left of Panel (a) displays how risk tolerance varies geographically in the U.S., using an incentivized elicitation. The map on the right of Panel (a) shows the geographic variation of the Preference Survey Module's (PSM; Falk et al., 2023) qualitative self-assessment of risk tolerance in our data.[3] The geographic patterns vary considerably between the two measures. For example, according to self-assessments, Upper Midwest residents appear to be highly risk-averse, while choices in incentivized elicitations suggest they are relatively risk-tolerant. This divergence is not unique to risk or to specific geographic units, as shown in Panel (b). The scatterplots in this panel compare U.S. states' rankings based on citizens' responses to traditional elicitations and qualitative self-assessments across the six domains of the PSM. As the plots reveal, the correlations between these rankings are low and statistically insignificant for all six domains. For instance, knowing that residents of a state rank highly in self-reported altruism tells us

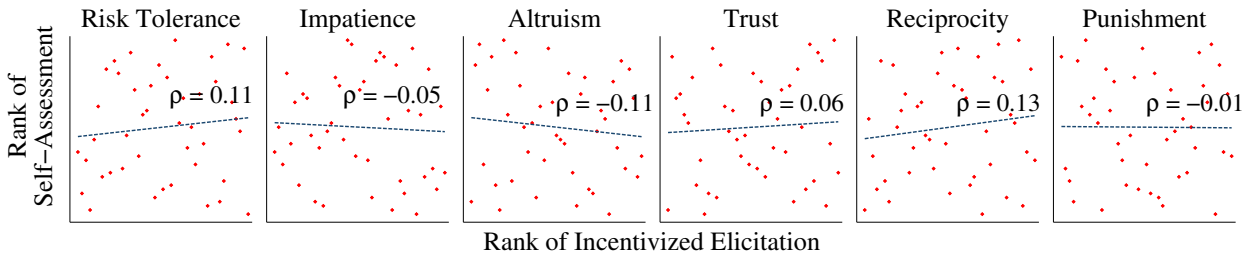
[3]To create these maps, we take the approach used by Falk et al. (2018) to plot the geographical distribution of preference indices, and apply it separately to incentivized elicitations and qualitative self-assessments. Specifically, we standardize each variable to a mean of 0 and a standard deviation of 1, then calculate the average of this standardized measure within each U.S. census division. Values below 0 indicate that the division's average is below the national mean. Appendix A includes similar maps to those in Panel (a) for impatience, trust, altruism, reciprocity, and punishment.

Figure 1: Qualitative self-assessments and incentivized elicitations identify different geographic distributions of risk preferences ($N = 4,950$).

(a) Estimated Risk Tolerance using Different Elicitations



(b) Ranking of preferences across U.S. states



Notes: The top panel of the figure plots the mean incentivized (left-hand side) and self-assessment (right-hand side) measures of risk tolerance for each census division, from online surveys of two representative samples of the U.S. population (U.S. Samples 1 and 2; see Table 2). The bottom panel uses the same data to plot the rank of the average self-assessment (y-axis) and incentivized measure (x-axis) at the state level. ρ is the Spearman rank correlation coefficient, and lines represent linear fits of the state-level data. States with fewer than 25 observations are excluded ($N = 43$).

little about how they would rank in actual giving behavior in a dictator game.

Why do two measures, which are highly correlated by economics standards, produce such divergent patterns? In Section 2, we show theoretically that even a very high correlation between two measures does not render them interchangeable.[4] Specifically, it does not guarantee that the correlation between one measure and an auxiliary variable—such as income,

[4]This is understood in econometrics, see McCallum (1972) and Wickens (1972). However, to the best of our knowledge, the literature on experimental validation has not drawn on these findings.

education, or geographical location—will coincide, in either magnitude or sign, with the correlation between another highly correlated measure and the same auxiliary variable. This is due to the fact that the signs of correlations are not transitive.[5] Indeed, we show that even a correlation as high as 0.9 between two variables does not preclude the possibility that these variables could exhibit opposite-signed correlations with other variables.

We find lower correlations between incentivized elicitations and qualitative self-assessments than those reported in the PSM, in line with previous replications in student samples (Vieider et al., 2015; Kosfeld et al., 2026), in Section 4. In contrast with earlier studies, we examine a broad range of populations, and correct for white noise measurement error in both the incentivized elicitations and qualitative self-assessments. Even with these corrections, the correlations we observe are on average around 2/3 of those of Falk et al. (2023). Further, the lower magnitudes of the correlations we find are not explained by particularly noisy responses in our data; for example, measurement error in the original data from the PSM is generally higher than in our samples.

In Section 5, we demonstrate that the correlations between qualitative self-assessments and auxiliary variables are driven primarily by variation in qualitative self-assessments that is unrelated to incentivized elicitations. First, we show that among the six qualitative self-assessments analyzed, four display stronger correlations with incentivized measures of constructs they were not designed to proxy for, even after correcting for measurement error, than with their incentivized counterpart. For example, the qualitative self-assessment of risk tolerance correlates more strongly with the incentivized altruism elicitation than with the incentivized risk elicitation. Second, we illustrate that while qualitative self-assessments are robustly correlated with a range of demographic, economic, and self-reported behavioral variables, these correlations are virtually unchanged even after controlling for all the incentivized elicitations. That is, the variation in qualitative self-assessments associated with

[5]For example, suppose $a = u + w$, $b = u + v$, and $c = v - w$, where u, v, and w are independent random variables. Then, $\text{Corr}[a, b] > 0$, $\text{Corr}[b, c] > 0$, while $\text{Corr}[a, c] < 0$.

these variables is largely unrelated to any incentivized elicitation.[6]

One argument in favor of qualitative self-assessments is their apparent simplicity: they ask participants to describe their own traits using colloquial language. However, in Section 6, we show that those with lower cognitive ability seem to rely more heavily on response heuristics. In particular, we document the frequent use of focal responses: 0, 5, or 10 on an 11-point scale. We find that participants with lower cognitive ability are more likely to give such responses than participants with higher cognitive ability or participants in student samples. Moreover, comparisons between U.S. and U.K. samples suggest that cultural differences influence how qualitative scales are interpreted. These findings indicate that inference from qualitative self-assessments may be confounded by respondents' cognitive and cultural characteristics, and challenge the idea that these questions are inherently simple to answer.

We discuss the implications of our findings for preference elicitation in Section 7. Our findings highlight the need for more rigorous approaches to validating new measures. Even when experimentally validated, qualitative self-assessments appear to capture information about individuals' traits that differs from what is elicited through the incentivized measures against which they are benchmarked. This information may reflect aspects of preferences that are of independent interest, but it may also arise from factors unrelated to preferences, such as self-perception, framing effects, or response heuristics.[7] Future research should therefore aim to identify more precisely what qualitative self-assessments measure, rather than treating experimental validation as a sufficient justification for their use.

[6]The qualitative self-assessments and incentivized elicitations yield quite different patterns of correlations between preferences and other individual characteristics; see Appendix Figure A.3. This is particularly true of correlations with cognitive ability, where self-assessments suggest that higher cognitive ability participants are less risk tolerant and more impatient, contrary to what previous literature suggests (see, for instance, Dohmen et al., 2010; Snowberg and Yariv, 2021).

[7]A literature in psychology documents the potential usefulness of qualitative self-assessments in certain contexts, particularly when multiple self-assessments are employed within the same domain. See, for example, Blais and Weber (2006), Frey et al. (2017), and Huang et al. (2024).

2 A Simple Theory of Experimental Validation

A simple theoretical model helps clarify the conditions under which experimentally validated proxies can produce reliable inferences. Experimental validation aims to find a proxy variable q (for qualitative self-assessment) for a preference variable p, which is measured by an incentivized elicitation. The goal is to use q to estimate the relationship between some y—for example, income, education, location, or responses to an experimental treatment—and p when it may be difficult to measure p directly. To simplify exposition, we assume that these variables have already been standardized, that is $\mathbb{E}[p] = \mathbb{E}[y] = \mathbb{E}[q] = 0$, and $\text{Var}[p] = \text{Var}[y] = \text{Var}[q] = 1$.[8]

Experimental validation ensures that q is (perhaps highly) positively correlated with p. Both p and q may be correlated with y. Thus, we model q as:

$$q = (1-\gamma)p + \gamma\eta = (1-\gamma)p + \gamma(\delta\eta_y + (1-\delta)\eta_p)$$

in which $\gamma, \delta \in [0,1]$ are unknown parameters. Here, η captures variation in q that is not directly linked to p. The noise term η can be decomposed into two components, η_p and η_y, with the property that $\text{Cov}[\eta_p, y] = \text{Cov}[\eta_y, p] = 0$. We assume that $\text{Cov}[\eta_y, y] \neq 0$. If $\delta > 0$, then inference will be confounded by η_y; hence, we label η_y a confounding factor. It is worth noting that q may encode additional information beyond p, some of which may be of substantive interest. Even so, this additional information can still confound inference. To illustrate, suppose that p is a measure of risk aversion, while η_y captures an individual's propensity to overweight small probabilities. In some applications, it may be immaterial which aspect of risk preferences is correlated with, say, intelligence. In many others, however, the implications differ markedly depending on whether q is interpreted as measuring risk aversion, probability weighting, or a combination of the two. Moreover, because experimental validation only suggests that q is associated with risk aversion, it provides no indication of

[8] This assumption is consistent with our empirical analysis, which uses standardized versions of continuous variables.

whether η_y reflects probability weighting, some other aspect of preferences, or an entirely different characteristic.

If $\delta = 0$, then η is white noise (with respect to y). Of course, this decomposition is specific to a particular variable y and may not generalize: η may represent white noise measurement error with respect to income, but not with respect to education. The possibility that $\text{Cov}[\eta_p, p] < 0$ allows for an experimental validation *understating* the amount of information about q found in p.

The model could be complicated to allow for additional noise from measurement, without changing its core insights. For simplicity, we assume that q and p (and also y) are observed perfectly: that is, η represents a genuine source of variation in q rather than imperfect measurement. Improving the accuracy of q through, for instance, developing an index of multiple measures will thus not remove issues related to the noise term η. Similarly, as highlighted by Figure 1, η may confound estimated population moments as well as individual-level estimates. Combining a qualitative self-assessment with another question type, as in the indices used by Falk et al. (2018, 2023), could, in principle, attenuate the impact of η somewhat, but would not eliminate it. In practice, as we discuss in Section 4, both q and p likely include classical measurement error. Consequently, in Section 5, we instrument p with a duplicate elicitation, following the approach of Gillen et al. (2019).

If both y and p are measured, we can obtain the desired regression coefficient:

$$y = \beta_p p + \varepsilon_1 \qquad \Rightarrow \qquad \beta_p = \text{Cov}[y, p]\,.^{9}$$

Estimating the same regression with the experimentally-validated proxy variable yields:

$$y = \beta_q q + \varepsilon_2 \qquad \Rightarrow \qquad \beta_q = (1-\gamma)\text{Cov}[y, p] + \gamma\delta\text{Cov}[\eta_y, y]\,.$$

[9] No constant is required in the regression, as $\mathbb{E}[y] = \mathbb{E}[p] = 0$ through our standardization. Moreover, variance does not appear in the denominator of the expression for the regression coefficient due to standardization; that is, $\beta_p = \text{Cov}[y, p] = \text{Corr}[y, p]$.

This equation shows both the potential and pitfalls of experimental validation.

Experimental validation produces a useful proxy when $\gamma\delta\text{Cov}[\eta_y, y] = 0$. If this is due to $\gamma = 0$, then $q = p$ and $\beta_q = \beta_p$. If, instead, $\gamma > 0$ and $\delta = 0$, then the noise in q is orthogonal to y and acts as white noise measurement error. This measurement error attenuates the estimated coefficient from the second regression so that $0 < |\beta_q| < |\beta_p|$. This attenuation can be corrected by using a second experimentally-validated variable q' as an instrument, as long as the noise in q' is orthogonal to y and to η.

When $\gamma\delta\text{Cov}[\eta_y, y] > 0$, experimental validation may produce a poor proxy: β_p and β_q may be of different signs, even if p and q are highly correlated. This can occur even if no part of the correlation between p and q is driven by η_p, namely when $\delta = 1$. For example, if $q = 0.9p + 0.1\eta_y$ then the correlation between p and q is 0.9. However, if $\text{Cov}[y, p] = \sqrt{1/20} \approx 0.22$ and $\text{Corr}[\eta_y, y] = -\sqrt{19/20} \approx -0.975$, then $\beta_q \approx -0.22 < 0 < 0.22 \approx \beta_p$.[10]

This simple framework demonstrates that even a very high correlation between an incentivized elicitation and a qualitative self-assessment in an experimental validation is insufficient to ensure that the latter is a valid proxy. The remainder of this paper presents evidence that these issues undermine inference when employing the most popular qualitative self-assessments. We first demonstrate, in Section 4, that 65–100% of each qualitative self-assessment we examine can be attributed to the noise term η.[11] Moreover, as we show in Section 5, the qualitative self-assessments we investigate are as, and sometimes more, highly correlated with other incentivized elicitations than with their target incentivized elicitation. Further, we show that correlations between qualitative self-assessments and variables of interest—income, cognitive ability, and so on—seem almost entirely due to the confounding factor η_y. Finally, in Section 6, we present evidence suggesting one possible source of the

[10]The magnitude of $\text{Corr}[\eta_y, y]$ in the example is the maximum consistent with $\text{Cov}[y, p] = \sqrt{1/20}$. Due to the normalization of q, $\text{Var}[\eta_y] = 19$, and so $\text{Cov}[\eta_y, y] = -\sqrt{19^2/20} \approx -4.25$. Assuming that p, y, and η_y are normally distributed, then, with only 300 observations, both $\widehat{\beta}_p$ and $\widehat{\beta}_q$ will be statistically significantly different than zero at the 0.05 level ∼97.5% of the time.

[11]The lower bound assumes that $\delta = 1$. If $\delta < 1$, then q may be entirely a combination of η_p and η_y, regardless of the correlation between p and q. For example, if $\text{Corr}[p, q] = x$, it can be rationalized by $\gamma = 1$, $\delta = 1 - x/z$, and $\text{Cov}[\eta_p, p] = z$, with $z \in [x, 1]$.

confounding factor. Specifically, we show that participants rely on heuristics when answering qualitative self-assessments. Lower cognitive ability participants are more likely to choose focal values at the extremes and middle of numerical response scales. Thus, responses to qualitative self-assessments may be correlated with cognitive ability for reasons unrelated to economic preferences. As cognitive ability is related to many economic outcomes, this creates a plausible pathway for spurious relationships between qualitative self-assessments and economic outcomes.

3 Measures and Datasets

Our data are drawn from a range of studies that examined economic preferences in various participant populations. Each study included both incentivized preference elicitations and qualitative self-assessments. They differ in the preference domains they considered. These studies were carried out between 2014 and 2021, and contain a total of 13,157 observations.

3.1 Measures

The central measures are qualitative self-assessments and the incentivized elicitations for which they are meant to proxy. We describe the most common way that the measures were implemented. Deviations from these descriptions are detailed dataset-by-dataset in the next subsection. Screenshots showing specific questions can be found in Appendix B.2.

Qualitative Self-Assessments: We examine qualitative self-assessments across the six domains studied in Falk et al. (2018) and Falk et al. (2023): risk tolerance (sometimes referred to simply as "risk"), impatience, altruism, trust, reciprocity, and punishment. Participants were asked to rate themselves on an 11-point scale, from 0 to 10, for each of these domains. Question wording can be found in Table 1. All the self-assessment measures were taken from an early draft of Falk et al. (2023), which was available in 2014 when we first included

Table 1: Qualitative Self-Assessments and Incentivized Elicitations

		PSM	GPS	GPS Survey
Risk Tolerance				
Qualitative:	How do you see yourself: are you a person who is generally willing to take risks or do you try to avoid taking risks?	✓	✓	✓
Incentivized:	Certainty equivalent of a lottery (as % of expected value)	✓		
Impatience				
Qualitative:	How well does the following statement describe you as a person? "I tend to postpone things even though it would be better to get them done right away."			✓
Incentivized:	Amount needed today to forgo a payment in the future	✓		
Altruism				
Qualitative:	How would you assess your willingness to share with others without expecting anything in return, for example your willingness to give to charity?	✓	✓	✓
Incentivized:	Amount sent in a dictator game (as % endowment)	✓		
Trust				
Qualitative:	As long as I am not convinced otherwise I always assume that people have only the best intentions.	✓	✓	✓
Incentivized:	Amount sent by first mover in a Trust Game (as % endowment)	✓		
Reciprocity				
Qualitative:	How would you assess your willingness to return a favor to a stranger?		✓	✓
Incentivized:	Amount returned by second mover in a trust game (as % received)	✓		
Punishment				
Qualitative:	Are you a person who is generally willing to punish unfair behavior even if this is costly?	✓	✓	✓
Incentivized:	Amount used to punish receiver returning 0 in trust game (as % endowment)			

Notes: PSM refers to Falk et al. (2023), GPS to Falk et al. (2018), and GPS Survey to the survey documents found at https://gps.iza.org/downloads. In contrast to our measure, the GPS measure of reciprocity does not specifically reference "a stranger."

qualitative self-assessments in our studies. Most questions from that early draft were later incorporated into the Global Preference Survey (GPS), with the exception of the punishment item, which was split into two parts.[12] The impatience question we use is included in the survey documents for the GPS, and perhaps the surveys themselves.[13]

Incentivized Elicitations: These measures are standard in the literature and very similar to those used by Falk et al. (2023), except for the punishment elicitation. Duplicate elicitations of risk and impatience were included in all datasets, and most datasets also included a single incentivized elicitation of the four social preferences: altruism, trust, reciprocity, and punishment. Table 1 provides brief descriptions of these elicitations, with screenshots in Appendix B.2. All variables are coded so that higher values consistently indicate greater levels of the targeted preference; for example, higher risk tolerance is reflected by either a higher self-assessed willingness to tolerate risk, or a higher certainty equivalent for a lottery.

For punishment, participants were presented with a trust game setting in which the sender sends the full amount, and the receiver returns nothing. Participants then decided how much of a given stock of experimental points they would like to spend to punish the receiver, with each point spent reducing the receiver's payoff by six points. In contrast, Falk et al. (2023), measured "negative reciprocity," in which punishment is inflicted on the opponent in either a prisoner's dilemma or an ultimatum game. The only other noteworthy difference is that, in their altruism measure, dictator giving benefited a charity, whereas ours directed the gift to another survey participant.

Cognitive Ability: Each survey measured participants' cognitive ability using a set of nine questions. Six questions were taken from the International Cognitive Ability Resource

[12]The GPS dataset has been used to address a range of questions, including the relationship between patience and economic development (Sunde et al., 2022), how economic preferences affect trade outcomes (Korff and Steffen, 2022), and gender gaps in preferences (Falk and Hermle, 2018).

[13] GPS survey documents were obtained from https://gps.iza.org/downloads. Specifically, the question appears to have been included as item WP13426, and described in the survey document for every language and country, but is not mentioned in either Falk et al. (2018) or the published version of Falk et al. (2023).

(ICAR, Condon and Revelle, 2014). Three of these questions are similar to Raven's Matrices, and the other three ask participants to determine which of several images displays a rotation of a given shape. The survey also contained the Cognitive Reflection Test (CRT; Frederick, 2005), which includes three arithmetic questions with an instinctive, but incorrect, answer. The cognitive ability score is the sum of correct answers to these nine questions.[14]

Individual Characteristics: Our representative samples contained measures of demographic and economic characteristics. We used sex, age, education, income, and a binary measure of whether an individual owns stocks or shares. In addition, we elicited self-reported measures of participants' health behaviors and other activities for a subset of respondents in U.S. Sample 2. We use this information in Appendix A.5.

3.2 Datasets

Table 2 provides high-level descriptions of the datasets we use. We refer readers to the papers that collected these datasets for details beyond what is essential for our discussions.

Falk et al. (2023): This dataset is the point of comparison for several analyses, and taken from the replication package of Falk et al. (2023). The 409 participants in this dataset, University of Bonn students who were recruited by the experimental laboratory, were divided into two groups of roughly equal size. The first group was initially presented with incentivized elicitations of risk and time and a battery of survey questions, including self-assessments and hypothetical incentivized questions about social preferences. One week later, the same group was presented with incentivized elicitations of social preferences and survey questions regarding risk and time preferences. The second group of participants completed these tasks in reverse order, also one week apart. Due to missing values, there are 360 observations for correlations in reciprocity and punishment, and 382 for correlations in other domains.

[14]Falk et al. (2018) use a qualitative self-assessment of mathematical ability as a proxy for cognitive ability. They ask participants to rate their agreement, from 0 to 10, with the statement "I am good at math."

Table 2: Datasets

			Main Measures	
	N	Year	Preferences (# duplicates)	Other
Falk et al. (2023) (Student)	409	2010–11	All Self-Assessments (1) Incentivized Risk, Time (2) Incentivized Social (2)	
U.S. Sample 1 (Representative)	1,950	2018–19	All Self-Assessments (2) Incentivized Risk, Time (2) Incentivized Social (2)	Demographics Cognitive Ability
U.S. Sample 2 (Representative)	3,000	2015–16	All Self-Assessments (1) Incentivized Risk, Time (2) Incentivized Social (1)	Demographics Cognitive Ability Behavior SA
U.K. Sample (Low-Education)	1,984	2017	All Self-Assessments (1) Incentivized Risk, Time (2) Incentivized Social (1)	Demographics
Caltech (Student)	3,266	2014–16	Risk, Time, Altruism SA (2) Incentivized Risk, Altruism, Time (2)	
Pitt (Student)	437	2021	Altruism, Trust, Reciprocity SA (1) Incentivized Risk (2)	
UBC (Student)	202	2019	Risk, Altruism SA (2) Incentivized Risk, Altruism (2)	
Mechanical Turk (Convenience)	2,318	2016, 2019	Risk, Altruism SA (2) Incentivized Risk, Altruism (2)	

Notes: SA stands for "self-assessment."

3.2.1 General Population Datasets

All of our general population samples, as well as the Pitt student sample introduced below, were surveyed online by YouGov, using a similar survey implementation. Appendix B.1 contains further details and screenshots.

U.S. Sample 1: This representative sample of the U.S. comes from Chapman et al. (2024). This is our main dataset as it repeatedly surveyed the same participants, making it possible to more closely mimic Falk et al. (2023). It also contained duplicate measures of all qualitative self-assessments and incentivized elicitations, allowing us to examine whether weak correlations are explained by classical measurement error.

U.S. Sample 2: There are two samples in this dataset. The first is a representative sample of the U.S. from Chapman et al. (2025a). The second is a representative sample of the U.S. from Chapman et al. (2023). These samples contain compatible versions of the measures used in this paper, so we combine them into a single dataset.

U.K.: This sample, from Barcellos et al. (2024), consists of low-education U.K. residents. Survey respondents are U.K. residents who had attended school in the U.K., left school at or before 16, and were born between September 1, 1954 and August 31, 1960.

3.2.2 Student and Convenience Sample Datasets

Caltech: The Caltech dataset combines the Fall 2014, Spring 2015, Fall 2015, and Spring 2016 waves of the incentivized surveys within the Caltech Cohort Study used in Gillen et al. (2019) and Jackson et al. (2026). It also includes the Caltech lab responses from Snowberg and Yariv (2021). Only the Spring 2016 wave includes a qualitative self-assessment of altruism ($N = 605$). Qualitative self-assessments of impatience do not appear in the Fall 2014 and Spring 2016 waves, leaving 1,778 observations for this measure. All impatience elicitations used hypothetical incentives. We treat all of these samples as a unified dataset, despite some repeated observations of individuals.

UBC: This lab sample of University of British Columbia students comes from Snowberg and Yariv (2021). The study itself is similar to those run within the Caltech Cohort Study.

Pitt: This is an incentivized survey of participants recruited from the Pittsburgh Experimental Economics Laboratory (PEEL) taken from Chapman et al. (2025b). The survey is similar to those in U.S. Sample 1 and 2.

Mechanical Turk: These two samples of Mechanical Turk workers are taken from Snowberg and Yariv (2021). The latter sample used incentive levels set at half those of the earlier sample, but the two are otherwise identical, and we analyze them as a combined dataset.

4 Experimental Validation in General Populations

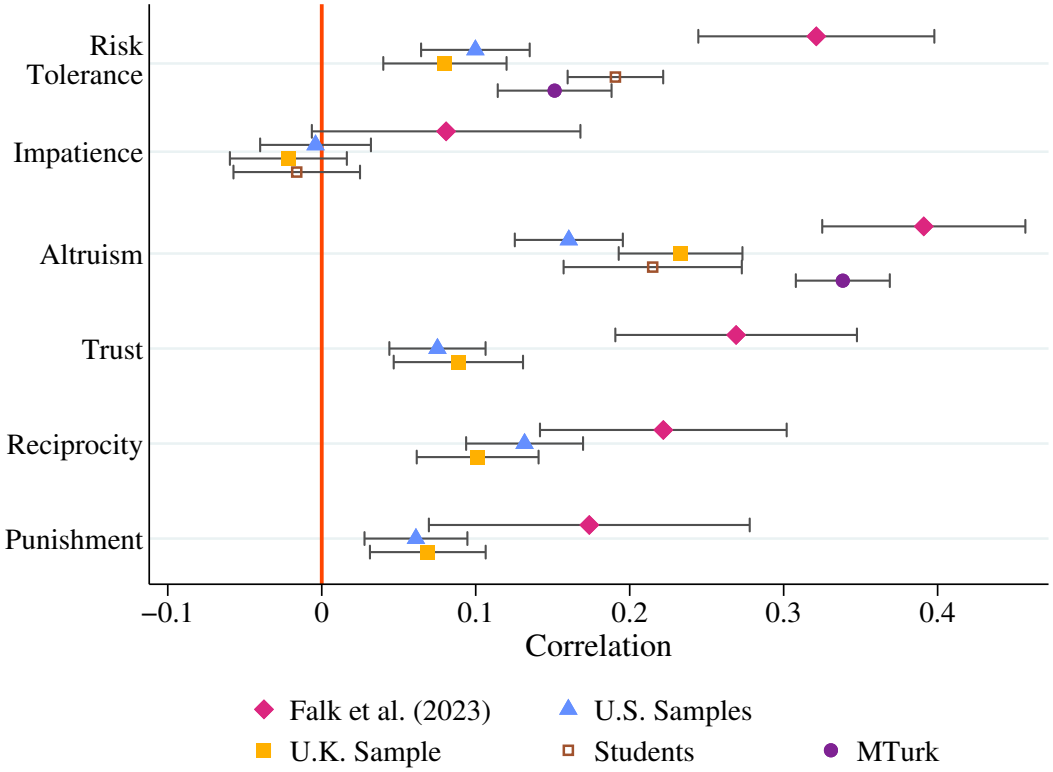
In line with previous experimental validation exercises, we find significant positive correlations between qualitative self-assessments and incentivized elicitations in five of six preference domains examined in the PSM/GPS. However, the magnitudes are consistently smaller than found in Falk et al. (2023). This pattern holds in all our datasets, across a range of subgroups within our representative samples, and does not appear to stem from measurement error.

Our datasets consistently reveal positive, yet modest, correlations between most qualitative self-assessments and incentivized elicitations, as shown in Figure 2. This implies that a relatively large share of the variation in qualitative self-assessments can be attributed to noise term η. In the language of the model presented in Section 2, $1 - \gamma = \text{Corr}[p, q]$, so the noise term's share of the variance in q is $1 - \text{Corr}[p, q]$, translating to roughly 60-90% for the Falk et al. (2023) data, and 80–100% in our U.S. samples. In fact, we observe smaller correlations than those in Falk et al. (2023) in each of our datasets. This suggests that modest correlations are not specific to general population samples.[15]

The smaller correlations we observe are unlikely to be explained by attenuation due to classical measurement error, as data from our samples are not unusually noisy. Table 3

[15]There is some evidence that correlations may be slightly higher in convenience samples than in broader samples, particularly for risk tolerance, but the observed differences are, in any case, small. Appendix A.2 shows that the magnitude of the correlations is similar when estimating Spearman correlations, as well as when using alternative incentivized elicitations, or when focusing on subgroups of our representative samples.

Figure 2: Qualitative self-assessments and incentivized elicitations exhibit low correlations in all datasets.



Notes: The figure compares correlations between qualitative self-assessments and incentivized elicitations across different samples. To ensure comparability, the correlations for each sample, except those for Falk et al. (2023), use a single self-assessment question, a corresponding incentivized elicitation for social preferences, and an average of two incentivized elicitations for risk tolerance and impatience. Correlations for Falk et al. (2023) also use the average of two incentivized elicitations for trust, reciprocity, and punishment. Bars represent 90% confidence intervals.

estimates the extent of classical measurement error by utilizing the duplicate measures of each preference available in our datasets.[16] As demonstrated in Gillen et al. (2019), the proportion of variation in a given measure within a dataset that is due to measurement error can be calculated as one minus the correlation between its duplicate measures. We report this proportion for different measures and samples in Table 3.[17]

The level of measurement error in U.S. Sample 1 compares favorably to those of North

[16]The duplicate incentivized elicitations differed in incentive levels and varied in other fine details. The duplicate qualitative self-assessments were identical, but asked at different points in the same survey.

[17]Formally, suppose X^a and X^b are two measures of the same underlying preference x^*. Classical measurement error implies that $X^a = X^* + \nu_X^a$ and $X^b = X^* + \nu_X^b$, with ν_X^a, ν_X^b i.i.d. random variables, and

American university students. In particular, U.S. Sample 1 has a lower level of measurement error than university students for the risk tolerance and impatience incentivized elicitations, and higher for the altruism incentivized elicitation.[18] Moreover, U.S. Sample 1 appears to have lower levels of measurement error than Falk et al. (2023) for qualitative questions, as shown in the last two columns of Table 3. U.S. Sample 1 repeats the same qualitative self-assessments about 20 minutes apart. Falk et al. (2023) ask nearly identical self-assessments in a single survey. From a large number of similar self-assessments, we selected a pair that seemed most alike, based on their variable labels. For example, for impatience, we use v158: "I often regret decisions that I make" and v162: "I make decisions I later regret."[19] Table 3 reveals another important pattern: qualitative self-assessments exhibit measurement error levels comparable to those of incentivized elicitations, challenging the notion that qualitative self-assessments are simpler to complete. We return to this point in Section 6.

The difference between our results and those of Falk et al. (2023) also does not appear to stem from differences in implementation, as evidenced in Figure 3. We mimic Falk et al.'s (2023) design using our U.S. Sample 1, the dataset most similar to theirs. Falk et al. (2023) collect responses to the same qualitative self-assessments and incentivized elicitations measured one week apart, with duplicate elicitations for five of the six incentivized elicita-

$\mathbb{E}\left[\nu_X^a \nu_X^b\right] = 0$. If we assume that $\frac{\text{Var}[\nu_X^a]}{\text{Var}[X^a]} = \frac{\text{Var}[\nu_X^b]}{\text{Var}[X^b]} \equiv \frac{\text{Var}[\nu_X]}{\text{Var}[X]}$, then

$$\widehat{\text{Corr}}[X^a, X^b] \to_p \text{Corr}[X^a, X^b] = \frac{\sigma^2_{X^*}}{\sigma^2_{X^*} + \sigma^2_{\nu_X}}.$$

Thus, $1 - \widehat{\text{Corr}}[X^a, X^b]$ captures the proportion of the measure's variation that is due to measurement error.

[18] Measurement error was similar in U.S. Sample 2—29% (s.e. = 2.2%) for risk tolerance, and 25% (3.0%) for impatience. Similarly, in the altruism domain, we observe measurement error in the qualitative self-assessments of 40% (1.7%) in the MTurk sample, and 44% (3.4%) among Caltech students. The duplicate qualitative self-assessment in these two samples asks if they would "go out of [their] way to do something nice for a stranger." The replication dataset of Falk et al. (2023) does not include the duplicate incentivized measures underlying the included average.

[19] For other domains, the questions are as follows. For risk, v104: "I like risky things," and v105: "I like taking risks." For altruism, v10: "Willingness to give to charities," and v12: "Willing to spend for charity even if I don[']t benefit." For trust: v72 "Willingness to trust," v83 "Compared to others, I easily trust people." For reciprocity, v53: "Willingness to reward a favor," and v64: "If someone does me a favor, I am willing to return it." For punishment, v36: "If someone puts me in a diff[icult] position, I'll do the same," and v38: "If someone intentionally harms me, I'll do the same to him/her."

Table 3: Measurement Error in Different Samples

	Incentivized Measures					Self-Assessments	
			Students				
	Sample 1	MTurk	Caltech	UBC	Pitt	Sample 1	Falk et al.
Risk Tolerance	25% (2.2)	35% (1.9)	26% (1.6)	31% (5.3)	29% (3.6)	18% (1.6)	35% (3.9)
Impatience	16% (1.9)		20% (2.7)			13% (1.4)	36% (4.0)
Altruism	44% (3.0)	20% (1.2)	14% (3.3)	26% (4.8)		27% (2.1)	35% (3.5)
Trust	39% (2.6)					18% (1.5)	35% (3.3)
Reciprocity	28% (1.9)					25% (3.2)	60% (4.8)
Punishment	33% (1.9)					33% (2.8)	28% (3.0)
N	1,950	2,318	3,001*	202	437	1,950	397†

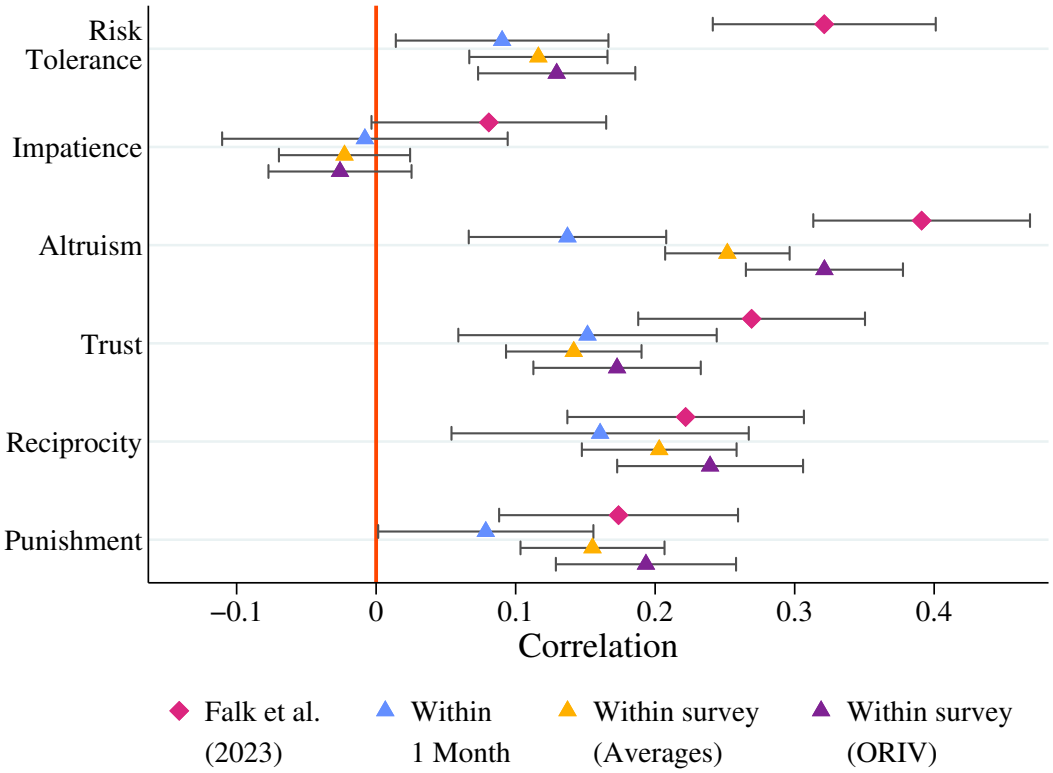
Notes: U.S. Sample 1 included two identical elicitations of each self-assessment. For Falk et al. (2023) we estimate measurement error using two self-assessments which appear most similar based on the variable labels in their replication dataset. *: $N = 3,001$ for risk tolerance, $N = 1,778$ for impatience, and $N = 605$ for altruism. †: the number of observations ranges from $N = 373$ (for reciprocity and punishment) to $N = 397$ (for trust and altruism). Bootstrapped standard errors in parentheses.

tions. We approximate this design by correlating the responses received in the initial survey in U.S. Sample 1, and those received at any point in the following five weeks, a total of 480 respondents.[20] Both the initial and follow-up surveys contained qualitative self-assessments and incentivized measures. Thus, each participant generates two data points, and we cluster standard errors at the individual level.[21] The correlations are markedly smaller than those reported by Falk et al. (2023) and, as might be expected, smaller than the within survey correlations displayed above. The impatience correlation is close to zero, and for punishment,

[20]U.S. Sample 1 is a multi-wave study in which the first survey had 1,950 participants, and then, each week, 150 randomly-chosen participants from the first survey were invited to complete the same survey.

[21]That is, we correlate qualitative self-assessments in the first survey with incentivized elicitations in the latter survey, and incentivized elicitations in the first survey with qualitative self-assessments in the latter survey, while clustering standard errors by individual.

Figure 3: Correlations are weak even after accounting for measurement error.



Notes: The "Within 1 month" series mimics the procedure used by Falk et al. (2023) as closely as possible in our datasets in U.S. Sample 1 ($N = 480$). As in Falk et al. (2023), this series uses one elicitation of each self-assessment, and, with the exception of altruism, the average of two elicitations for each incentivized measure. The "Within Survey (Averages)" and "Within Survey (ORIV)" series present correlations within the initial survey from U.S. Sample 1, first by averaging measures, and then using the ORIV technique to correct for measurement error. Bars represent 90% confidence intervals.

it is statistically indistinguishable from zero at the 5% level.

Figure 3 also shows that the correlations between qualitative self-assessments and incentivized elicitations remain modest even after accounting for possible attenuation due to classical measurement error. This figure includes, as the final set of estimates, the correlations derived by instrumenting one duplicate with the other, using ORIV (Gillen et al., 2019). This approach eliminates the attenuating effects of classical measurement error on correlations. Correspondingly, for altruism, reciprocity, and punishment, our estimated coefficients move substantially closer to those found in Falk et al. (2023). This is not the case, however, for risk, time, and trust. This comparison likely underestimates the discrepancy

between our results and those of Falk et al. (2023), as their data does not allow for a similar correction for measurement error.[22] More importantly, the magnitude of correlations implies that, at best, only one-third of the variation in qualitative self-assessments contains information about the preferences that they are meant to proxy for.

In sum, this section broadly replicates the experimental validation of Falk et al. (2023), and other papers, within a general population sample. Except for patience, we observe modest, positive correlations between qualitative self-assessments and incentivized elicitations in both general populations and convenience samples. However, in contrast to those earlier studies, we demonstrate that modest correlations are not explained by classical measurement error. This poses a major threat to inference, as most of the variation in qualitative self-assessments is not explained by choices in the corresponding incentivized elicitation.

5 Experimentally-Validated Measures May Not be Valid

Across all six domains, qualitative self-assessments are correlated with multiple incentivized elicitations. In only two of these domains is the largest correlation with the analogous incentivized elicitation. In addition, qualitative self-assessments are robustly correlated with demographic and economic variables in our data, as in some prior studies (for example, Dohmen et al., 2011). Yet, these correlations are not explained by the variation in a qualitative self-assessment associated with *any* of the six incentivized elicitations. Thus, it is challenging to interpret the correlations between qualitative self-assessments and demographic

[22]There is a countervailing force: the correlations observed between qualitative self-assessments and incentivized elicitations in Falk et al. (2023) are potentially inflated by their question selection procedure. Specifically, their method selects the linear combination of survey items that best predicts choices in incentivized elicitations. This implies that the reported correlations are conditional on having been ranked highly within their particular sample. In Appendix A.3, we use the Falk et al. (2023) replication dataset to simulate a similar procedure. We randomly split their sample into two subsamples and, for each subsample, select the qualitative self-assessments with the highest correlation with an incentivized elicitation. We then compare the selected questions and correlations across subsamples, and across 10,000 replications. Across the six domains, the same question is selected in both subsamples in 41% of replications, and the average Kendall's τ, a rank-order correlation, is 0.25. The estimated correlation is inflated by 0.06, which is 23% of the correlation in the full sample. Question selection is most unstable for punishment: the same question is chosen in both subsamples in fewer than 1% of replications, and the rank correlation is -0.03. It is most stable for reciprocity, for which the corresponding figures are 88% and 0.33.

Table 4: Correlations between Qualitative Self-Assessments and Incentivized Elicitations, using ORIV

		Incentivized Elicitations					
		Risk Tolerance	Impatience	Altruism	Trust	Reciprocity	Punishment
Qualitative Self-Assessment	Risk Tolerance	0.13*** (.034)	0.01 (.034)	0.19*** (.038)	0.15*** (.036)	0.07* (.035)	0.14*** (.032)
	Impatience	−0.01 (.033)	−0.03 (.031)	0.11*** (.036)	0.12*** (.035)	0.01 (.031)	0.02 (.033)
	Altruism	−0.01 (.037)	−0.09** (.036)	0.32*** (.035)	0.25*** (.036)	0.24*** (.038)	0.07** (.033)
	Trust	0.07** (.031)	0.02 (.032)	0.21*** (.037)	0.17*** (.037)	0.08** (.033)	0.01 (.033)
	Reciprocity	−0.00 (.039)	−0.09** (.036)	0.30*** (.035)	0.29*** (.036)	0.24*** (.040)	0.12*** (.035)
	Punishment	0.02 (.036)	−0.03 (.036)	0.10** (.041)	0.08** (.038)	0.08** (.035)	0.19*** (.040)

Notes: Data from U.S. Study 1, Week 0 ($N = 1,950$). Correlations are using ORIV, with bootstrapped standard errors. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

or economic variables.

5.1 Strong Correlations with Multiple Incentivized Elicitations

Qualitative self-assessments are often related to multiple incentivized behaviors, including many that are theoretically orthogonal to the concept they are trying to capture, as shown in Table 4. This table displays the complete ORIV correlation table between all incentivized elicitations and qualitative self-assessments in U.S. Sample 1. As can be seen, the qualitative self-assessment of risk tolerance is more highly correlated with the incentivized elicitations of altruism, trust, and punishment than with the incentivized elicitation of risk tolerance. Similar patterns hold for other qualitative self-assessments. Perhaps most disturbing, the

Table 5: Correlations between Incentivized Elicitations, using ORIV

	Risk Tolerance	Impatience	Altruism	Trust	Reciprocity	Punishment
Risk Tolerance		-0.15^{***} (.038)	0.05 (.041)	0.07^{*} (.042)	-0.00 (.040)	0.04 (.035)
Impatience	-0.15^{***} (.037)		-0.13^{***} (.039)	-0.17^{***} (.037)	-0.11^{***} (.037)	-0.07^{**} (.034)
Altruism	0.05 (.041)	-0.13^{***} (.040)		$1.02^{***,\dagger}$ (.030)	0.54^{***} (.040)	0.19^{***} (.042)
Trust	0.07^{*} (.041)	-0.17^{***} (.038)	$1.02^{***,\dagger}$ (.030)		0.67^{***} (.034)	0.18^{***} (.040)
Reciprocity	-0.00 (.040)	-0.11^{***} (.038)	0.54^{***} (.040)	0.67^{***} (.034)		0.22^{***} (.035)
Punishment	0.04 (.036)	-0.07^{**} (.034)	0.19^{***} (.041)	0.18^{***} (.040)	0.22^{***} (.035)	

Notes: Data from U.S. Study 1, Week 0 ($N = 1,950$). Correlations are using ORIV, with bootstrapped standard errors. ***, **, * denote statistical significance at the 1%, 5%, and 10% level. † unlike a standard correlation coefficient, correlations estimated by ORIV do not have an upper bound of 1.

qualitative self-assessment of impatience is slightly negatively correlated with the incentivized elicitation for impatience, but statistically significantly correlated with altruism and trust. The table suggests that the confounding factor η_y, introduced in Section 2, does indeed contain additional information on preferences. However, these preferences may be in a theoretically distinct domain: for example, in the case of impatience, η_y may capture altruism and trust.[23]

The correlations displayed in Table 4 do not reflect the correlation structure between incentivized elicitations, as shown in Table 5. This table displays ORIV correlations within the set of incentivized elicitations in U.S. Sample 1.[24] The patterns are substantially dif-

[23]The experimental data of Falk et al. (2023) also exhibit considerable variation in correlations across domains. We provide an analogue of Table 4 using the Falk et al. (2023) data in Appendix Table A.2.

[24]Since the same elicitations appear on both axes, the table is symmetric. We report correlations both

ferent. For example, the incentivized risk elicitation is not significantly correlated with the incentivized altruism elicitation, whereas the qualitative self-assessment of risk is most strongly correlated with the incentivized altruism elicitation. In Appendix Table A.4 we also show that correlations between qualitative self-assessments display different patterns than the cross-correlations. Taken together, these observations indicate that the patterns in Table 4 are not due to inherent linkages between incentivized elicitations.

5.2 Associations with Demographic and Economic Characteristics

Nearly all of the variation in qualitative self-assessments associated with demographic and economic characteristics is independent of the variation in incentivized elicitations. To interpret the findings that demonstrate this, it is helpful to revisit our model from Section 2. In that section, p denoted an incentivized elicitation, y a demographic (or other) variable. We modeled qualitative self-assessments as $q = (1-\gamma)p + \gamma\eta$, in which the noise term $\eta = \delta\eta_y + (1-\delta)\eta_p$ is decomposed such that $\text{Cov}[\eta_p, y] = \text{Cov}[\eta_y, p] = 0$.[25]

The validity of q as a proxy for p cannot be assessed by comparing β_p, the coefficient from a regression of y on p, to β_q, the coefficient from a regression of y on q. In particular, even if $\beta_q = \beta_p$, it may still be the case that β_q is entirely driven by the confounding factor η_y. For example, suppose that $\gamma = 1$ and $\text{Cov}[\eta_y, y] = \beta_p/\delta$. In this case, $\beta_q = \beta_p$, despite the fact that q contains none of the relevant variation in p. Thus, $\beta_q = \beta_p$ for one auxiliary variable y would not translate to other variables of interest.

To evaluate the validity of q (with respect to y), the following regression is helpful:

$$q = \beta_{y|p}y + \beta_{p|y}p + \varepsilon_3 \qquad \Rightarrow \qquad \beta_{y|p} = \frac{\gamma(\delta\text{Cov}[\eta_y, y] - (1-\delta)\beta_p\text{Cov}[\eta_p, p])}{1-\beta_p^2},$$

in which the expression for $\beta_{y|p}$ follows from an application of the Frisch–Waugh–Lovell

below and above the diagonal to facilitate comparison with Table 4.

[25] This decomposition is generally not unique. Hence, it is not possible to identify δ, η_p, and η_y without further assumptions. In any decomposition, however, $\text{Cov}[\eta_i, i] \neq 0 \iff \text{Cov}[\eta, i] \neq 0$, for $i \in \{p, y\}$.

Table 6: Relationships between qualitative self-assessments and demographics are unrelated to variation in incentivized elicitations (N = 1,950).

	Dependent Variable = Qualitative Self-Assessment					
	Risk Tolerance			Altruism		
Cognitive Ability	-0.12^{***}	-0.13^{***}	-0.12^{***}	-0.04	-0.06^{**}	-0.06^{**}
	(.028)	(.028)	(.030)	(.028)	(.027)	(.030)
Male	0.34^{***}	0.35^{***}	0.33^{***}	-0.20^{***}	-0.22^{***}	-0.21^{***}
	(.055)	(.054)	(.055)	(.056)	(.055)	(.054)
Age	-0.09^{***}	-0.10^{***}	-0.10^{***}	0.17^{***}	0.17^{***}	0.16^{***}
	(.028)	(.028)	(.027)	(.028)	(.028)	(.027)
Education	0.05	0.05	0.06^{**}	0.03	0.03	0.03
	(.031)	(.031)	(.031)	(.029)	(.029)	(.029)
Income (Log)	0.00	0.01	0.00	-0.02	-0.02	-0.01
	(.035)	(.035)	(.035)	(.034)	(.033)	(.033)
Stock Investor	0.07	0.07	0.08	0.06	0.05	0.04
	(.065)	(.064)	(.065)	(.061)	(.062)	(.061)
Incentivized Elicitation:						
Risk Tolerance		0.15^{***}	0.14^{***}			-0.05
		(.037)	(.037)			(.039)
Impatience			0.04			-0.06
			(.035)			(.037)
Altruism			0.29^{**}		0.35^{***}	0.46^{***}
			(.139)		(.048)	(.130)
Trust			-0.06			-0.20
			(.159)			(.145)
Reciprocity			-0.04			0.11^{**}
			(.058)			(.055)
Punishment			0.12^{***}			-0.02
			(.037)			(.037)

Notes: All columns use data from U.S. Sample 1. Incentivized elicitations are instrumented to eliminate the effect of classical measurement error. Coefficients and standard errors (in parentheses) on all continuous measures are standardized. All specifications include an indicator variable for missing income. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

theorem. If there is only white noise error in q (with respect to both y and p)—that is, if $\text{Cov}[\eta_y, y] = \text{Cov}[\eta_p, p] = 0$—then $\beta_{y|p} = 0$. We use this insight to evaluate qualitative self-assessments vis à vis a variety of demographic and economic variables.

The variation in qualitative self-assessments and demographic and economic characteristics appears to be independent of the variation in qualitative self-assessments that is associated with incentivized elicitation(s), as shown in Table 6. This table regresses the qualitative self-assessments of risk tolerance and altruism on cognitive ability, gender, age, education, income, and an indicator for stock ownership.[26] The first column for each measure shows that these self-assessments are generally strongly correlated with demographic and economic variables. The second column for each measure introduces the incentivized measure as a control. As shown above, if qualitative self-assessments differ from their counterparts only in the addition of white noise measurement error, the coefficients on demographic and economic variables would go to zero. However, the coefficients are largely unchanged, indicating that almost *none* of the correlations between self-assessments and demographics or economic variables are driven by variation in the incentivized elicitations. This observation is not due to measurement error in the incentivized elicitations: we use the duplicate incentivized elicitations as instruments for one another, ensuring that attenuation bias in the coefficients does not allow the demographic and economic variables to absorb excess variation (see Table 3, and surrounding text, in Gillen et al., 2019). Further, this pattern is in spite of the fact that the incentivized elicitations themselves are often statistically significantly correlated with the demographic and economic variables (see Appendix Figure A.3).

We find similar results when examining the relationship between qualitative self-assessments and self-reported behaviors, including health behaviors, community engagement, and political interest; see Appendix A.5. Consistent with previous studies, we find that qualitative self-assessments, as well as incentivized elicitations, are correlated with a range of self-reported behaviors, although the interpretation of these correlations is not always clear. For instance, self-assessed risk tolerance is positively correlated with both unhealthy behaviors, like smoking and binge drinking, and healthy behaviors, like exercise and eating fruit and vegetables. These correlations are not driven by the variation in incentivized elicitations.

[26] Appendix A.4 shows similar patterns for impatience, trust, reciprocity, and punishment. For risk tolerance and impatience, we can also incorporate data from U.S. Sample 2, and obtain similar results.

Further, controlling for self-reported behaviors has little effect on estimated correlations between demographics and qualitative self-assessments. Together, these findings suggest that qualitative self-assessments are capturing variation unrelated to preferences they purport to represent.

5.3 Bounds on Correlations Driven by the Confounding Factor

Most of the correlations between qualitative self-assessments and demographics stem from the confounding factor η_y. To demonstrate this, we derive estimable bounds on the share of these correlations attributable to the confounding factor.

We restate two regression specifications discussed in Section 2, with two additional specifications that are useful for further analysis:

$$
\begin{aligned}
y = \beta_p p + \varepsilon_1 \qquad &\Rightarrow \qquad \beta_p = \text{Cov}[y, p] \\
q = \beta_{qp} p + \varepsilon_2 \qquad &\Rightarrow \qquad \beta_{qp} = (1-\gamma) + \gamma(1-\delta)\text{Cov}[\eta_p, p] \\
y = \beta_q q + \varepsilon_3 \qquad &\Rightarrow \qquad \beta_q = (1-\gamma)\beta_p + \gamma\delta\text{Cov}[\eta_y, y] \\
y = \beta_{q|p} q + \beta_{p|q} p + \varepsilon_4 \qquad &\Rightarrow \qquad \beta_{q|p} = \frac{\gamma\delta\text{Cov}[\eta_y, y] - \gamma(1-\delta)\beta_p\text{Cov}[\eta_p, p]}{1-\beta_{qp}^2}
\end{aligned}
$$

Similar to our discussion of $\beta_{y|p}$ above, if η is pure white noise error (with respect to both y and p), so that $\delta = 0 = \text{Cov}[\eta_p, p]$, then $\beta q|p = 0$.

When $\delta \neq 0$, some of the correlation between q and y stems from the confounding factor. We now bound the segment of β_q that is driven by correlation that is attributable to the confounding factor, namely $\Psi \equiv \gamma\delta\text{Cov}[\eta_y, y]$.

We can use the equations above to write:

$$
\begin{aligned}
\beta_{q|p} &= \frac{\Psi - \gamma(1-\delta)\beta_p\text{Cov}[\eta_p, p]}{1-\beta_{qp}^2}, \\
\Psi &= \beta_{q|p}(1-\beta_{qp}^2) + \gamma(1-\delta)\beta_p\text{Cov}[\eta_p, p]\,.
\end{aligned}
$$

We now derive lower and upper bounds on Ψ. For both, it is useful to note that, by construction, the variance of both η_p and η_y are bounded by 1. It follows that $(1-\delta)|\text{Cov}[\eta_p, p]\,| \leq (1-\delta)\sqrt{\text{Var}[\eta_p]} \leq 1$. This is a direct consequence of the correlation between η_p and p falling in the $[-1, 1]$ interval. We assume that $\beta_p \geq 0$. Analogous arguments follow for $\beta_p < 0$.

Since $\beta_p \geq 0$, the segment Ψ is bounded above by β_q. Furthermore, since $(1-\delta)|\text{Cov}[\eta_p, p]\,| \leq 1$, Ψ is maximized when $\beta_{qp} = \gamma(1-\delta)\text{Cov}[\eta_p, p]$. In other words, Ψ is maximized when $\gamma = 1$, implying that $\Psi = \beta_q$. Thus, we can never rule out the possibility that the correlation between y and q is entirely due to confounding factors.

The minimum value of Ψ is achieved when $\gamma(1-\delta)\text{Cov}[\eta_p, p]$ is minimized (given β_{qp} and $\beta_{q|p}$). Since $|\text{Cov}[\eta_p, p]\,| \leq \sqrt{\text{Var}[\eta_p]}$, the variance of η_p must satisfy

$$\text{Var}[q] = 1 = (1-\gamma)^2 + \gamma^2(1-\delta)^2\text{Var}[\eta_p] + \gamma^2\delta^2\text{Var}[\eta_y]\,.$$

The value of $\text{Var}[\eta_p]$ is maximized when $\text{Var}[\eta_y] = 0$. As this implies that η_y is a constant, this is also consistent with $\delta\text{Cov}[\eta_y, y] = 0$. In this case, $|\gamma(1-\delta)\text{Cov}[\eta_p]\,| \leq \sqrt{1-(1-\gamma)^2}$. In particular, it follows that $\gamma(1-\delta)\text{Cov}[\eta_p] \geq -\sqrt{1-(1-\gamma)^2}$. To get a lower bound on Ψ, we therefore need to minimize $-\sqrt{1-(1-\gamma)^2}$, subject to $\gamma, \delta \in [0, 1]$ and $\beta_{qp} = (1-\gamma) - \sqrt{1-(1-\gamma)^2}$. That minimum is achieved at $-\frac{1}{\sqrt{2}}\sqrt{1-\beta_{qp}\sqrt{(2-\beta_{qp}^2)}}$.

Using symmetric arguments for $\beta_p < 0$, we have the following:

Proposition (Bounds on Correlation Segment due to the Confounding Factor)**.** *The segment Ψ of the coefficient β_q that is due to the confounding factor η_y satisfies*

$$\mathcal{B}^- = \beta_{q|p}(1-\beta_{qp}^2) - \beta_p\tilde{\beta}_{qp} \leq \Psi \leq \beta_q = \mathcal{B}^+ \qquad \text{if } \beta_p \geq 0$$
$$\mathcal{B}^+ = \beta_q \leq \Psi \leq \beta_{q|p}(1-\beta_{qp}^2) - \beta_p\tilde{\beta}_{qp} = \mathcal{B}^- \qquad \text{if } \beta_p < 0,$$

in which $\tilde{\beta}_{qp} = \dfrac{1}{\sqrt{2}}\sqrt{1-\beta_{qp}\sqrt{(2-\beta_{qp}^2)}}$.

Assuming again that $\beta_p \geq 0$, a few implications follow. First, an estimated $\beta_{q|p} = 0$ does

not ensure that $\Psi \approx 0$. Indeed, the regression estimation corresponding to $\beta_{q|p}$ does not allow us to rule out the possibility that the segment Ψ that is due to the confounding factor is "canceled out" by attenuation in the estimated coefficient due to white noise. Nevertheless, if $\beta_{q|p} = 0$, then the interval specified in the proposition will always contain 0. That is, zero correlation due to the confounding factor ($\Psi = 0$) cannot be ruled out. As such, checking whether $\beta_{q|p} = 0$ can provide a helpful (minimal) test for future experimental validations. Second, if β_p and β_q have different signs, then we can be sure that $|\Psi| > 0$. Third, if $\beta_{qp} = 1$ then $\Psi \in [0, \beta_q]$. That is, even if p and q are perfectly correlated, we cannot rule out the possibility that β_q is driven entirely by the confounding factor.

Corollary (Minimum Segment of β_q due to the Confounding Factor)**.** *The minimum segment of the coefficient β_q that is driven by confounding factors satisfies:*

$$\Psi \geq \begin{cases} 0 & \text{if } 0 \in [\mathcal{B}^-, \mathcal{B}^+] \\ \min\{|\mathcal{B}^-|, |\mathcal{B}^+|\} & \text{otherwise.} \end{cases}$$

Table 7 displays the portion of the coefficients β_q that remain after removing Ψ, the minimum portion that is explained by the confounding factor using the corollary. Nearly all the coefficients are close to zero, and few are statistically significant. That is, even in the best case scenario, nearly all the predictive power of these measures is due to the confounding factor η_y.

6 Response Heuristics in Qualitative Self-Assessments

Earlier sections highlight limitations of qualitative self-assessments as proxies for traditional economic preference measures. However, their use could also be justified by their perceived simplicity. For example, Dohmen et al. (2018, p. 126) contend that the "simplicity of [the] general risk question....has the advantage of being easy to understand, thereby limiting the problem of decision errors or noise." We have already shown in Table 3 that qualitative self-

Table 7: After accounting for confounding factors, qualitative self-assessments provide limited evidence that preferences are correlated with economic or demographic characteristics.

	Cognitive Ability	Male	Age	Income (Log)	Education	Stock Investor
Risk	−0.01	0.00	0.00	0.00	0.00	0.00
Tolerance	(.032)	(.031)	(.030)	(.037)	(.034)	(.031)
Impatience	0.00	0.01	−0.06*	−0.04	−0.04	−0.00
	(.031)	(.031)	(.032)	(.033)	(.031)	(.032)
Altruism	0.00	0.00	0.02	0.01	0.01	0.05
	(.033)	(.034)	(.033)	(.039)	(.035)	(.034)
Trust	0.00	0.00	0.00	0.04	0.02	0.09***
	(.030)	(.032)	(.031)	(.033)	(.032)	(.031)
Reciprocity	0.09***	0.03	0.07**	0.03	0.02	0.05
	(.033)	(.035)	(.035)	(.039)	(.035)	(.034)
Punishment	0.00	0.05	0.00	0.04	0.00	0.00
	(.037)	(.037)	(.041)	(.044)	(.040)	(.039)

Notes: $N = 1,950$. The table displays univariate correlations between qualitative self-assessments and individual characteristics (β_q), using the corollary to remove the minimum portion that is explained by confounding factors. Incentivized elicitations and qualitative self-assessments are instrumented to eliminate the effect of classical measurement error. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

assessments, are, if anything, more noisy than incentivized elicitations. The analyses in this section reveal that, additionally, the use of response heuristics in qualitative self-assessments is correlated with cognitive ability, challenging the notion that they are inherently straightforward for participants. Moreover, we provide evidence of cross-cultural variation in how self-assessments are interpreted. Both of these factors pose problems for valid inference.

A simple example illustrates how qualitative self-assessments may be as, if not more, difficult to answer than more objective measures. Imagine visiting an optometrist and being asked to rate your eyesight on a scale from 0 to 10. Now, consider being asked to assess your skills as an economist using the same scale. How would you interpret the meaning of each number in these contexts? Would you be confident that your family and/or colleagues would use the same interpretation? Would assigning a score be easier than taking a vision test or collecting a citation count?

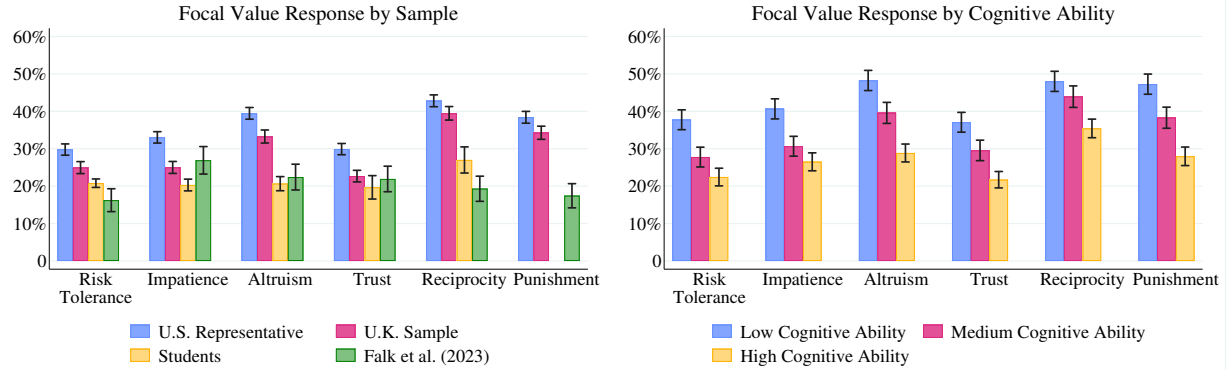
The qualitative self-assessments we study likely pose similar challenges to participants, but with added conceptual complexity. To respond carefully to "How willing are you to take risks, in general?", for example, a participant should i) determine the conceptual content of the question: for instance, which types of risks are being referred to; ii) consider how that concept applies to their own behavior and experiences; and then iii) decide how to aggregate that information and project it onto an 11-point numerical scale. Consequently, while responses to self-assessments might result from preferences, they could also reflect a combination of participants' reference groups, self-perceptions, social expectations, and their understanding of numerical scales, much of which may vary independently of preferences.[27] The coefficients relating to self-assessments in Table 6 may be capturing exactly such factors.

Qualitative self-assessments appear more challenging to answer for those with lower cognitive ability, as evidenced by Figure 4. This figure shows the rate of focal value response (FVR; Chapman et al., 2025a)—defined as selecting 0, 5, or 10—across different samples (left panel) and among cognitive ability terciles within our representative samples of the U.S. (right panel).[28] FVR levels are notably higher in broad population samples than in student samples, whether from the U.S. or from Germany (using data from Falk et al., 2023). This disparity appears to stem from participants in the representative sample who rank in the bottom tercile of cognitive ability. The pattern is consistent with lower cognitive ability individuals "rounding" their responses when faced with the perceived complexity of translating their preferences into an 11-point response scale (Barrington-Leigh, 2024). The relationship between FVR and cognitive ability also suggests that heuristics may be one source of the confounding factor η_y leading to the spurious correlations between qualitative self-assessments and other individual characteristics documented in the previous section.

[27] Arslan et al. (2020) report that individuals, when completing the risk qualitative self-assessment, think about behaviors they engage in. See Krosnick and Presser (2010) for a general discussion of theoretical challenges associated with numerical rating scales.

[28] Other studies have observed higher levels of FVR in general populations than we do. In the GPS, 40% of responses to the trust qualitative self-assessment were at focal values, compared to 30% in our representative samples of the U.S. (based on the replication dataset of Falk et al. (2018)). Similarly, Dohmen et al. (2011, see Figure 1 in) report high FVR rates for the risk tolerance self-assessment.

Figure 4: Focal Value Response (FVR) is more common in the general population, particularly among participants with low cognitive ability.

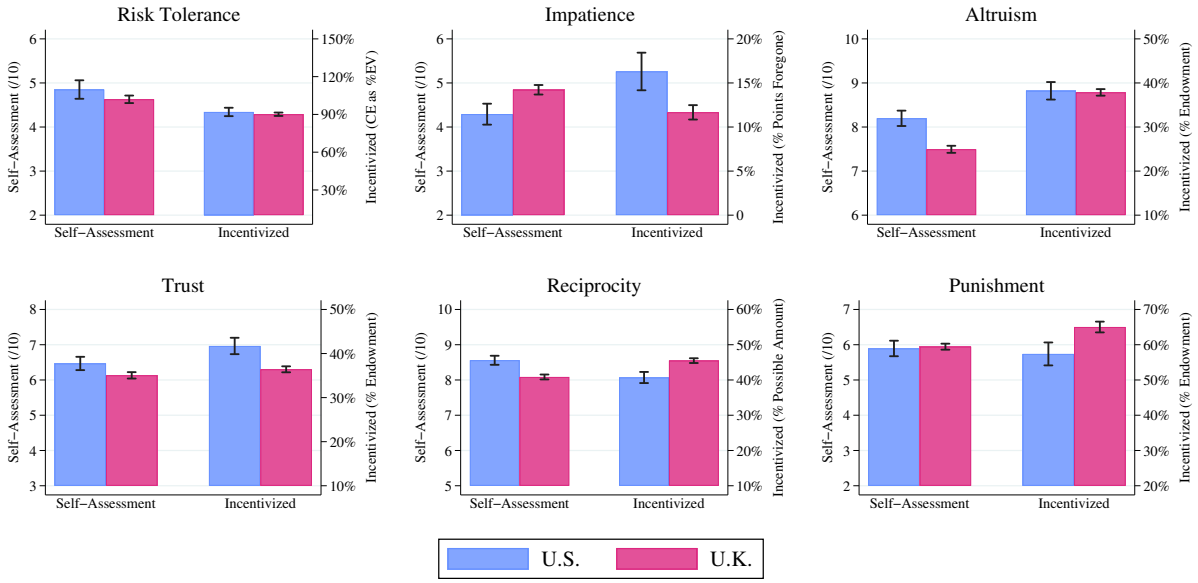


Notes: The figure displays the percentage of responses that were focal values—0, 5, or 10—in each qualitative self-assessment. U.S. Representative includes U.S. Samples 1 and 2 (combined $N = 4,950$). Students combines Caltech, Pittsburgh, and UBC samples.

The observed pattern of FVR implies that qualitative self-assessments suffer from similar measurement issues as established incentivized elicitations, negating a purported advantage of the method (Eckel, 2019). Chapman et al. (2025a; Figure 8) report comparable levels and patterns of FVR for both qualitative self-assessments and multiple price lists (MPLs) used to elicit risk tolerance and impatience. For example, Chapman et al. (2025a) report that for MPLs measuring risk tolerance, 41% of participants with low cognitive ability gave focal value responses, compared to 31% and 30% for participants with medium and high cognitive ability.

The differential use of heuristics such as FVR is particularly troubling as, unlike classical measurement error, they may lead to biased estimates, rather than simple attenuation. For example, suppose individuals with lower cognitive ability are more likely to choose 0 than those with higher cognitive ability, while all cognitive-ability groups exhibit similar propensities to choose 5 or 10 on the Impatience QSA (as appears to be the case; see Appendix Figure A.8). In this case, lower cognitive ability would appear to be associated with greater patience, as reported in Appendix Table A.8. This pattern runs counter to the relationship between patience and cognitive ability observed using the incentivized elicitation (see Appendix Figure A.3) and that observed in prior literature (see Dohmen et al. 2010).

Figure 5: Cultural differences affect responses to qualitative self-assessments.



Notes: The figure compares the average preference within different samples. The U.K. sample includes all participants in the U.K. low education sample ($N = 1,984$). These individuals are contrasted with a comparable subsample of U.S. Samples 1 and 2: those aged over 55 and with at most high school education ($N = 803$). Conclusions are similar when comparing to the entire U.S. Sample, see Appendix Figure A.4.

A more detailed comparison of our U.K. and U.S. samples, displayed in Figure 5, suggests that cultural background may play a role in respondents' interpretation of qualitative self-assessments.[29] This figure compares the average responses to qualitative self-assessments and incentivized elicitations in our sample of older, low-education, adults in the U.K., to those from a comparable subsample of U.S. participants. While cultural differences affect responses to both types of elicitations, they are at least as pronounced for qualitative self-assessments. Furthermore, in line with our discussion in previous sections, among the six domains, only trust exhibits differences between the U.S. and the U.K. that are in the same direction (and statistically significant) for both the qualitative self-assessment and the incentivized elicitation. The differences are particularly striking for impatience and reciprocity.

[29]Differences in interpretation across groups, within or between countries, would indicate that qualitative self-assessments lack measurement invariance, and preclude meaningful comparisons between those groups. See Mellenbergh (1989) and the more recent survey by Dong and Dumas (2020).

7 Discussion

We examine both the experimental validation method and the validity of qualitative self-assessments for core economic preferences, using data from over 13,000 participants in a number of representative, convenience, and student samples. Across these samples, correlations between qualitative self-assessments and incentivized elicitations are consistently modest, even after adjusting for measurement error. Moreover, multiple qualitative self-assessments exhibit stronger correlations with incentivized measures of other constructs. Notably, the links between various attributes—geography, demographics, and behaviors—and qualitative self-assessments differ from the links between these same attributes and incentivized elicitations. Experimental validation is thus insufficient to ensure valid inference in broad populations. Finally, while qualitative self-assessments are often promoted as simpler than incentivized elicitations, our findings indicate that low cognitive-ability participants disproportionately select salient options, such as the extremes or midpoints of numerical scales.

In what follows, we discuss the implications of our findings for the use of qualitative self-assessments and for preference elicitation. This discussion rests on the premise that incentivized choice-based elicitations are a widely used benchmark for the measurement of economic preferences (see Snowberg and Yariv, eds, 2025). Incentivized elicitations are grounded in the revealed preference paradigm, creating a direct connection between what is elicited and the underlying models they inform. For instance, Falk et al. (2023, p. 1946) emphasize that "the guiding methodology for developing the [preference survey] modules is identifying survey items that can predict well the choices in incentivized experiments." Measures that do not reflect choices in incentivized experiments may nevertheless be of value. Our work highlights scope for the development of richer qualitative measures, and the need for deeper understanding of the constructs captured by qualitative self-assessments.

Implications for Findings Based on Self-Assessments: Our findings call into question the conclusions drawn from studies that interpret qualitative self-assessments as measures

of specific preferences. Our results are consistent with previous research indicating that qualitative self-assessments may capture differences in understanding, perceptions, habits, or even other aspects of preferences (Paulhus and Vazire, 2007; Brenner and DeLamater, 2016; Arslan et al., 2020; Schneider and Sutter, 2026). As such, considering qualitative self-assessments as substitutes for the incentivized elicitations used to validate them may yield misleading conclusions. When both qualitative and choice-based responses are available, it may be informative to analyze the relationship between the two, and to understand when the different measures generate disparate findings. For example, making responses to the individual questions underlying the GPS indices accessible to the research community, separating responses to the qualitative self-assessments from those to quantitative questions, would facilitate this process and enhance credibility.[30]

Implications for Preference Elicitation: The appeal of qualitative self-assessments largely stems from their perceived ability to facilitate preference measurement across broad populations. Our results suggest that these measures do not accurately capture preferences reflected in incentivized elicitations. Qualitative self-assessments may still be informative about how individuals perceive, say, the riskiness of their behavior (Schonberg et al., 2011), or how they conceptualize their "ideal self" (Brenner and DeLamater, 2016). While these perceptions may be important in their own right, they may bear little connection to the preferences economists typically study or to the utility parameters that incentivized elicitations are designed to capture. Future work could develop models that align with these constructs, and then design new qualitative self-assessments explicitly around those models.

There may be scope to develop alternatives to qualitative self-assessments that mitigate some of the issues identified in this paper. Expanding qualitative measures into longer,

[30]Trust and negative reciprocity in the GPS are assessed solely through qualitative self-assessments. Measures for risk, impatience, and altruism include experimental tasks with hypothetical incentives, while reciprocity is evaluated using a hypothetical scenario involving a "thank you gift." Falk et al. (2018) combine the multiple questions for each preference into a single index, and so their replication dataset does not include responses to the underlying questions. The authors of Falk et al. (2018) have not been willing to share the disaggregated responses.

more nuanced question sequences could help disentangle preferences from behaviors, perceptions, and differences in interpretation. The Domain-Specific Risk-Taking (DOSPERT) scale, for example, employs 30 questions that focus primarily on behaviors, rather than self-assessments (Blais and Weber, 2006). Alternative paths could pursue further exploration of whether monetary incentives are essential for traditional choice-based elicitations (Stango and Zinman, forthcoming; Brañas-Garza et al., 2023), or develop tools to address issues associated with numerical response scales (Benjamin et al., 2023).

Our findings suggest that experimental validation is likely to be challenging in practice. The bounding exercise developed in Section 5.3 can be used to evaluate new or existing measures. However, such an exercise validates a measure only for the specific independent variables and population under study. As a result, it does not identify a measure that is portable across settings or research questions. In many cases, it may therefore be more straightforward to elicit the benchmark measure than seek experimentally validated alternatives. That said, conducting a validation exercise within a subsample of the broader population may still be valuable prior to including such measures in a large-scale survey.[31]

Implications for Measurement: Qualitative self-assessments have been endorsed on the basis of *predictive validity*, that is, the extent to which they predict auxiliary variables, behaviors, or outcomes (Schildberg-Hörisch, 2018). This criterion is susceptible to many of the concerns raised in this paper. First, predictive validity alone does not rule out endogeneity. For example, a qualitative self-assessment of risk attitudes may predict stock market participation extremely well, not because it measures risk attitudes accurately, but because the question inadvertently prompts respondents to think about their own investment behavior. In this case, the risk attitudes measure subsumes a measure of investment behavior; in our framework, η_y effectively reflects y.

[31] Dohmen et al. (2011) provide an example of such an approach, in carrying out an experimental validation in a representative sample of 450 participants, before studying a qualitative self-assessment from a survey of 22,000 participants. See Snowberg and Yariv (2025) for a systematic discussion of criteria for evaluating new measures.

Second, assessing a measure based on predictive validity can reinforce misspecified models. Continuing the example above, suppose stock market participation is driven by cognitive ability rather than risk attitudes, and, as we find in Section 6, the qualitative self-assessment of risk attitudes has some variation that is driven by cognitive ability (in η_y). A researcher who believes that risk attitudes drive investment may incorrectly conclude that the measure is a superior proxy for risk because it predicts investment behavior more accurately. Such reasoning can ultimately lead to misleading scientific conclusions and policy recommendations.

We therefore believe that new measures should satisfy stronger criteria than predictive validity alone. At a minimum, they should possess *construct validity*, in the sense of being grounded in a falsifiable theoretical framework. They should further exhibit *directional responsiveness*, namely empirical comparative statics consistent with that framework. For further discussion, see Snowberg and Yariv (2025).

Conclusion: Incentivized preference elicitation has become increasingly feasible due to advances in methodology and the growth of online survey platforms. While incentivized elicitations have been criticized for their limited predictive power, this concern has been mitigated by improved techniques for addressing measurement error (Beauchamp et al., 2017; Gillen et al., 2019; Chapman et al., 2025b; Jagelka, 2024). For instance, the incentivized elicitations in this paper demonstrate meaningful correlations with a range of individual characteristics (Appendix Figure A.3) and self-reported behaviors (Appendix Figure A.6). Despite these advances, there remains room to improve incentivized elicitations; see, for example, Chapman et al. (2025a) and Gerhardt and Suchy (2024).[32] We hope that future research will focus on improvements to theoretically motivated choice-based elicitations.

[32]Chapman et al. (2025a) use the Dynamically Optimized Sequential Experimentation (DOSE) method to elicit risk and time preferences in a representative online survey. FVR rates in DOSE stand at around 6%, with minimal variation across cognitive ability levels, suggesting the method is simple to understand.

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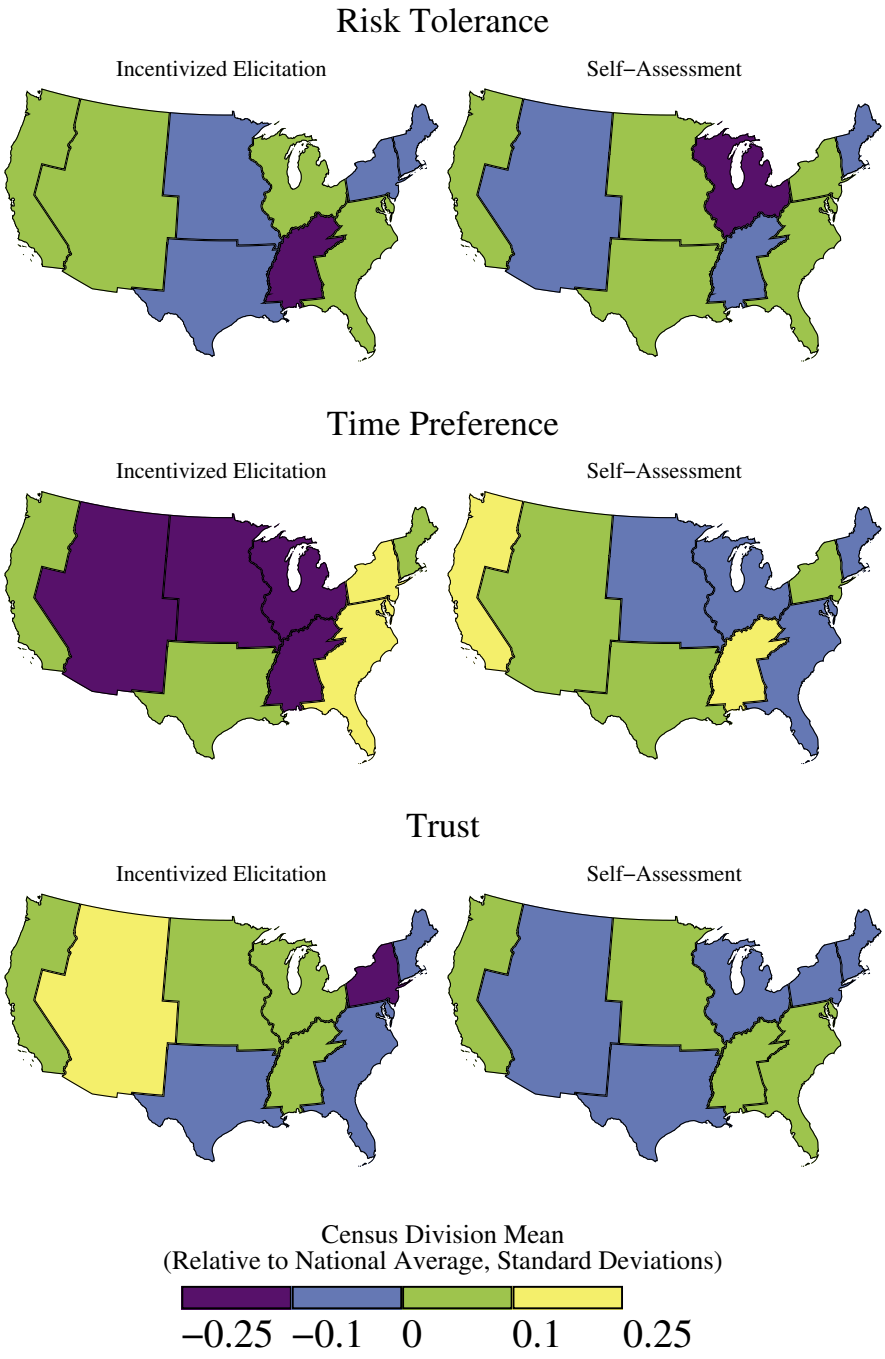
Online Appendix—Not Intended for Publication

A Additional Results

A.1 Distribution of Preferences within the United States

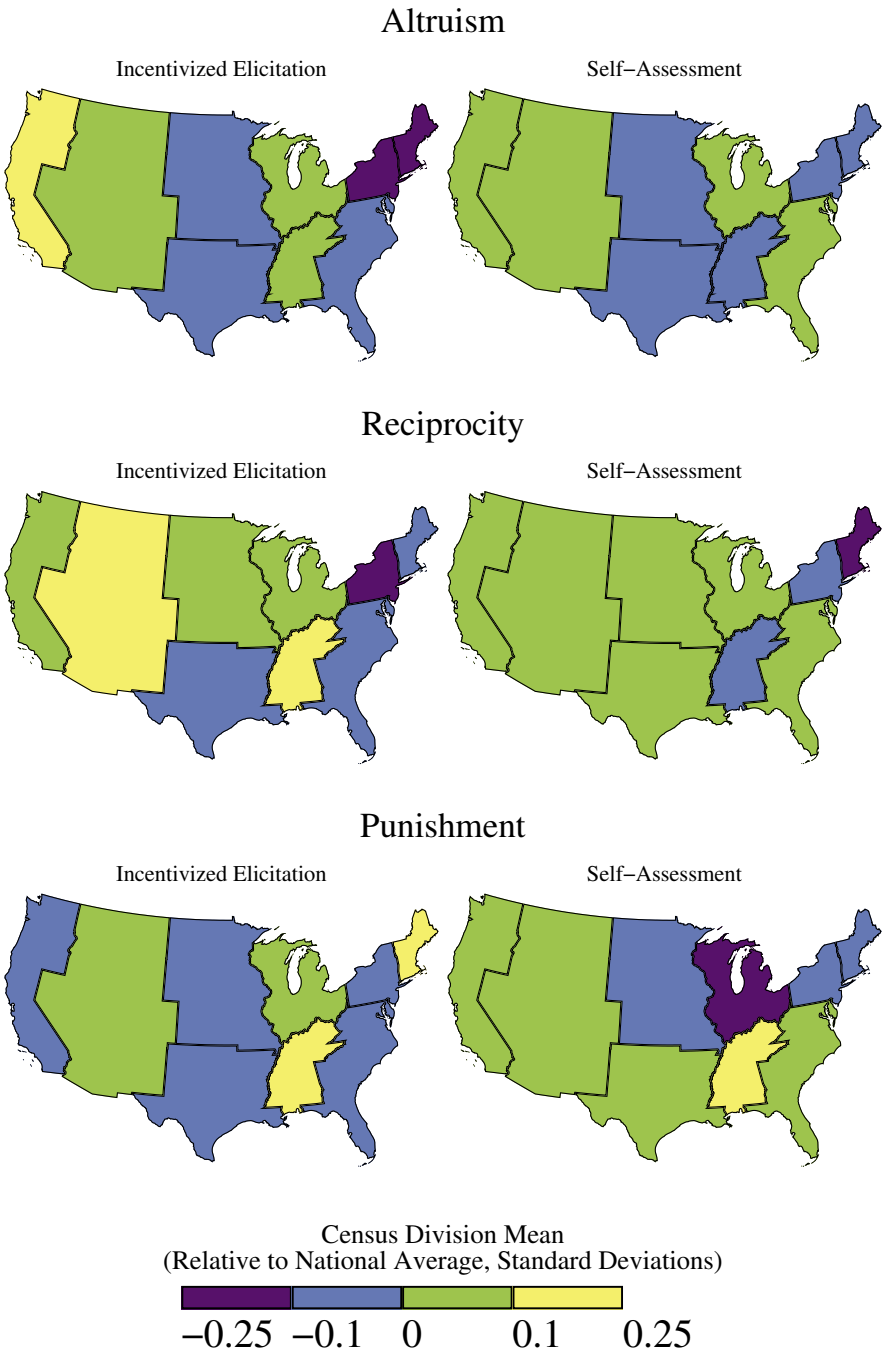
Figures A.1 and A.2 generalize Figure 1 in the main text and display the distribution of each preference at the census division level. The left-hand side of each figure displays incentivized elicitations, while the right-hand side displays the corresponding qualitative self-assessments. In each case, there are marked differences between patterns corresponding to qualitative self-assessments and the respective incentivized elicitations.

Figure A.1: Risk tolerance, time preferences, and trust in the United States.



Notes: Data from combined U.S. Samples 1 and 2. Alaska and Hawaii are included in the Pacific division. Each variable is standardized, then collapsed to state level. The scale is in standard deviations, relative to a mean of 0.

Figure A.2: Altruism, reciprocity, and punishment in the United States.



Notes: Data from combined U.S. Samples 1 and 2. Alaska and Hawaii are included in the Pacific division. Each variable is standardized, then collapsed to state level. The scale is in standard deviations, relative to a mean of 0.

A.2 Additional Correlations Between Qualitative Self-Assessments and Incentivized Elicitations

Table A.1 displays correlations between qualitative self-assessments and alternative incentivized elicitations of risk tolerance and punishment within U.S. Sample 1. Our preferred measure of risk tolerance elicited a participant's Willingness-to-Accept payment for a lottery. This measure has the virtue of being similar to Falk et al. (2023)'s measure, and also being available in U.S. Sample 2 and the U.K. Sample. The survey also included a number of other elicitations of risk preferences, including participants' Willingness-to-Pay for a lottery, their Certainty Equivalent for two lotteries, and their Certainty Equivalent for a draw from a risky urn and a draw from an ambiguous urn. For punishment, our preferred measure involves punishing the person that sent nothing back in a trust game, after receiving the maximum possible amount from the sender. We also elicited participants' willingness to engage in "anti-social punishment"—punishing the person that sent the full amount in the same trust game (who then received nothing in return). Each of these measures was elicited twice. As can be seen, the correlations between our preferred measures and these alternative elicitations are of a similar magnitude.

Table A.2 presents the correlations between qualitative self-assessments and incentivized elicitations using the replication dataset from Falk et al. (2023). As in our representative sample (Table 4), qualitative self-assessments are often statistically significantly correlated with incentivized elicitations other than those they are intended to proxy for. Table A.3 compares the correlations in Falk et al. (2023) to those in the subsample of our data that most closely resembles Falk et al. (2023)'s design—participants in U.S. Study 1 that completed two surveys within one month (see Figure 3 and surrounding discussion). Both Spearman and Pearson correlations are significantly smaller in our data than in Falk et al. (2023). Table A.4 complements Table 4 and 5 in the main text. The table displays ORIV correlations between qualitative self-assessments in U.S. Sample 1. This table echoes the message that the patterns in Table 4 are not due to inherent linkages between attributes.

Table A.1: Correlations with Alternative Elicitations

	Within	Within Survey	
	1 Month	Averages	ORIV
	Correlations with Risk Self-Assessment		
Preferred Measure	0.09*	0.12***	0.13***
	(.046)	(.030)	(.034)
Willingness To Pay	0.13**	0.17***	0.19***
	(.051)	(.031)	(.035)
Risk Aversion Urn	0.09*	0.10***	0.11***
	(.046)	(.031)	(.034)
Ambiguous Urn	0.15***	0.17***	0.19***
	(.044)	(.030)	(.032)
	Correlations with Punishment Self-Assessment		
Preferred Measure	0.08*	0.07**	0.09**
	(.047)	(.031)	(.038)
Anti-Social Punishment	0.09*	0.12***	0.13***
	(.046)	(.030)	(.034)
N	480	1,950	1,950

Notes: Data from U.S. Sample 1. Bootstrapped standard errors in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

Tables A.5 and A.6 present correlations between qualitative self-assessments and incentivized elicitations within various sub-groups of our U.S. general population samples. In each case, we use one qualitative self-assessment, the average of two incentivized elicitations for risk tolerance and impatience, and a single incentivized elicitation of social preferences.

Table A.2: Correlations between Qualitative Self-Assessments and Incentivized Elicitations in Falk et al. (2023)

		Incentivized Elicitations					
		Risk Tolerance	Impatience	Altruism	Trust	Reciprocity	Punishment
Qualitative Self-Assessment	Risk Tolerance	0.32*** (.047)	−0.02 (.052)	−0.03 (.051)	0.09* (.049)	−0.10** (.052)	0.03 (.052)
	Impatience	0.09* (.051)	0.08 (.053)	0.08 (.049)	0.06 (.048)	−0.00 (.050)	0.01 (.047)
	Altruism	−0.02 (.054)	0.13** (.052)	0.39*** (.039)	0.16*** (.052)	0.24*** (.051)	−0.01 (.062)
	Trust	0.03 (.046)	0.19*** (.051)	0.17*** (.050)	0.27*** (.048)	0.23*** (.050)	−0.09* (.052)
	Reciprocity	−0.03 (.052)	0.12** (.050)	0.11** (.052)	0.23*** (.049)	0.22*** (.049)	−0.05 (.055)
	Punishment	0.05 (.053)	−0.00 (.052)	0.00 (.057)	0.05 (.059)	0.15*** (.053)	0.17*** (.062)

Notes: Data from Falk et al. (2023), $N = 360$ for measures of reciprocity and punishment, and $N = 382$ for all other domains. Correlations are Pearson correlations, with bootstrapped standard errors. ***, **, * denote statistical significance at the 1%, 5%, and 10% level. Color increases in intensity at each 0.05 of magnitude.

Table A.3: Comparison of Pearson and Spearman Correlations

	Correlation Between Self-Assessment and Incentivized Elicitation			
	Pearson		Spearman	
	Replication	Falk et al. (2023)	Replication	Falk et al. (2023)
Risk Tolerance	0.09^{*}	0.32^{***}	0.10^{**}	0.35^{***}
	(.046)	(.047)	(.043)	(.046)
Impatience	-0.01	0.08	0.03	0.05
	(.062)	(.053)	(.052)	(.052)
Altruism	0.14^{***}	0.39^{***}	0.13	0.38^{***}
	(.043)	(.039)	(.045)	(.044)
Trust	0.15^{***}	0.27^{***}	0.13^{**}	0.28^{***}
	(.056)	(.047)	(.054)	(.048)
Reciprocity	0.16^{**}	0.22^{***}	0.13^{**}	0.21^{***}
	(.065)	(.049)	(.053)	(.049)
Punishment	0.08^{*}	0.17^{***}	0.08	0.16^{***}
	(.047)	(.063)	(.049)	(.053)

Notes: Replication refers to correlations within one month, as in Figure 3 ($N = 480$). Estimates for Falk et al. (2023) are based on their replication dataset. Bootstrapped standard errors in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

Table A.4: Correlations between Qualitative Self-Assessments, using ORIV

	Risk Tolerance	Impatience	Altruism	Trust	Reciprocity	Punishment
Risk Tolerance		0.16^{***} (.035)	0.19^{***} (.034)	0.27^{***} (.037)	0.20^{***} (.035)	0.30^{***} (.037)
Impatience	0.16^{***} (.035)		-0.03 (.033)	0.06^{*} (.034)	-0.02 (.032)	0.17^{***} (.036)
Altruism	0.19^{***} (.034)	-0.03 (.033)		0.44^{***} (.028)	0.72^{***} (.033)	0.15^{***} (.036)
Trust	0.27^{***} (.036)	0.06^{*} (.034)	0.44^{***} (.028)		0.39^{***} (.032)	0.14^{***} (.037)
Reciprocity	0.20^{***} (.035)	-0.02 (.032)	0.72^{***} (.033)	0.39^{***} (.032)		0.22^{***} (.037)
Punishment	0.30^{***} (.036)	0.17^{***} (.037)	0.15^{***} (.036)	0.14^{***} (.038)	0.22^{***} (.038)	

Notes: Data from U.S. Study 1, Week 0 ($N = 1,950$). Correlations are using ORIV, with bootstrapped standard errors. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

Table A.5: Correlations between Qualitative Self-Assessments and Incentivized Elicitations, by Subgroup

	Risk	Impatience	Altruism	Trust	Reciprocity	Punishment
All	0.10***	−0.00	0.16***	0.08***	0.13***	0.06***
	(.022)	(.022)	(.021)	(.019)	(.023)	(.020)
			$N = 4{,}950$			
ICAR: Above Median	0.16***	−0.04	0.16***	0.08***	0.15***	0.12***
	(.026)	(.029)	(.029)	(.026)	(.029)	(.025)
			$N = 2{,}816$			
ICAR: Top $\sim$ 10%	0.16***	0.01	0.26***	0.16***	0.18***	0.18***
	(.056)	(.061)	(.056)	(.061)	(.054)	(.054)
			$N = 561$			
ICAR: Top $\sim$ 5%	0.20**	0.12	0.38***	0.17**	0.15**	0.26***
	(.098)	(.090)	(.073)	(.074)	(.075)	(.067)
			$N = 235$			
CRT: No Questions Correct	0.07**	0.01	0.14***	0.06**	0.11***	0.05*
	(.029)	(.028)	(.028)	(.026)	(.032)	(.027)
			$N = 2{,}725$			
CRT: One or More Questions Correct	0.15***	−0.00	0.19***	0.11***	0.17***	0.10***
	(.029)	(.035)	(.030)	(.029)	(.029)	(.029)
			$N = 2{,}225$			
CRT: All Three Questions Correct	0.18***	−0.06	0.22***	0.18***	0.17***	0.05
	(.053)	(.083)	(.058)	(.055)	(.049)	(.071)
			$N = 500$			
High School or Less	0.08**	0.04	0.15***	0.09***	0.14***	0.04
	(.038)	(.037)	(.034)	(.031)	(.040)	(.035)
			$N = 1{,}689$			
Some College or College Degree	0.12***	−0.06**	0.18***	0.08***	0.13***	0.07***
	(.025)	(.027)	(.028)	(.024)	(.028)	(.025)
			$N = 2{,}690$			
Advanced Degree	0.12**	0.08	0.08	−0.03	0.10*	0.19***
	(.054)	(.054)	(.074)	(.071)	(.059)	(.049)
			$N = 571$			
Response Time: Not Fastest 10%	0.11***	−0.02	0.16***	0.07***	0.13***	0.06***
	(.023)	(.023)	(.022)	(.020)	(.023)	(.021)
			$N = 4{,}504$			
Response Time: Not Slowest or Fastest 25%	0.11***	−0.05	0.15***	0.05**	0.11***	0.07***
	(.030)	(.033)	(.030)	(.025)	(.031)	(.028)
			$N = 2{,}495$			

Notes: Data from combined U.S. Samples 1 and 2 ($N = 4{,}950$). Bootstrapped standard errors in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

Table A.6: Correlations between Qualitative Self-Assessments and Incentivized Elicitations, by Subgroup (Continued)

	Risk	Impatience	Altruism	Trust	Reciprocity	Punishment
All	0.10***	−0.00	0.16***	0.08***	0.13***	0.06***
	(.022)	(.022)	(.021)	(.019)	(.023)	(.020)
				$N = 4{,}950$		
Female	0.10***	−0.02	0.11***	0.06***	0.11***	0.04
	(.029)	(.028)	(.030)	(.025)	(.027)	(.026)
				$N = 2{,}688$		
Male	0.11***	0.02	0.21***	0.09***	0.15***	0.08**
	(.032)	(.034)	(.029)	(.029)	(.037)	(.032)
				$N = 2{,}262$		
Investor	0.10***	−0.03	0.09**	0.03	0.10***	0.08**
	(.034)	(.035)	(.039)	(.032)	(.036)	(.033)
				$N = 1{,}720$		
Not Investor	0.10***	0.00	0.19***	0.09***	0.14***	0.06**
	(.027)	(.027)	(.024)	(.023)	(.029)	(.025)
				$N = 3{,}230$		
Age: Under 40	0.06	0.03	0.20***	0.06*	0.10**	0.07*
	(.038)	(.039)	(.033)	(.033)	(.042)	(.038)
				$N = 1{,}694$		
Age: 40 to 60	0.14***	−0.06*	0.10**	0.10***	0.18***	0.06*
	(.035)	(.036)	(.038)	(.033)	(.032)	(.033)
				$N = 1{,}682$		
Age: Over 60	0.11***	−0.02	0.16***	0.07**	0.10***	0.05*
	(.036)	(.029)	(.038)	(.029)	(.034)	(.032)
				$N = 1{,}574$		
Above Median Income	0.11***	−0.05	0.13***	0.05*	0.13***	0.05*
	(.030)	(.029)	(.030)	(.029)	(.026)	(.027)
				$N = 2{,}491$		
Above 90% Income	0.15***	0.00	0.03	−0.00	0.17***	0.06
	(.054)	(.059)	(.071)	(.059)	(.048)	(.047)
				$N = 646$		
Above 95% Income	0.10	0.03	−0.06	−0.08	0.17***	0.09
	(.063)	(.041)	(.086)	(.077)	(.065)	(.064)
				$N = 395$		

Notes: Data from combined U.S. Samples 1 and 2 ($N = 4{,}950$). Bootstrapped standard errors in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

A.3 Simulated Question Selection Procedure Using Experimental Validation

In this subsection, we simulate the use of experimental validation to select qualitative self-assessments, using the Falk et al. (2023) replication dataset. We investigate only the qualitative self-assessments in their data, excluding quantitative and hypothetical questions. For each of 10,000 replications, we randomly divide the Falk et al. (2023) dataset into two groups, a and b. We first identify in group a the qualitative self-assessment with the highest absolute correlation with the incentivized elicitation: the question q_a that would be chosen by experimental validation within group a. We then compute the difference in the correlation between the incentivized elicitation, p, and q_a across the two groups: $D_a = \rho^a(q_a, p) - \rho^b(q_a, p)$, where ρ^g is the correlation computed within group $g \in \{a, b\}$. We then repeat the same procedure to compute D_b. We then compute the average of D_a and D_b.

Table A.7 documents the results. The first column reports the number of candidate qualitative self-assessments included in Falk et al. (2023) for each domain. The second column presents the correlation for the highest ranked qualitative self-assessment in the full sample. Columns three and four report the estimated difference in the correlation due to the question selection procedure, the mean of D_a and D_b, both in levels and as a percentage of the correlation in column 2. The final two columns report the stability of the question ranking, both in terms of identifying the highest ranked question (column 5) and the ranking of all questions (column 6).

The simulation identifies considerable heterogeneity across the preference domains. Altruism and reciprocity appear least affected by bias or instability in question selection: the estimated upward bias is close to 0, and the same question is chosen in both subsamples in more than 80% of replications. The punishment domain is highly affected, with upward bias of 0.14, and the same question selected in both subsamples in less than 1% of replications. This result reflects the fact that several questions in the full sample have very similar correlations with the incentivized elicitation.

Table A.7: Simulation of Selection Question Procedure Based on Experimental Validation

	Full Sample		Simulation			
	Num.	Best	Δ Correlation		Rank Stability	
	QSAs	Corr.	Level	% of Best	% Top	Corr.
Risk Tolerance	48	0.32	0.05	15%	47%	0.21
Impatience	41	0.39	0.05	12%	10%	0.41
Altruism	19	0.39	0.01	3%	84%	0.25
Trust	20	0.32	0.07	21%	18%	0.35
Reciprocity	11	0.35	0.02	5%	88%	0.33
Punishment	21	0.17	0.14	80%	0%	−0.03

Notes: The table reports results from a simulation of an experimental validation question selection procedure, using the replication dataset of Falk et al. (2023). The first two columns relate to the full dataset, and report the number of qualitative self-assessments, and the best correlation between a self-assessment and the incentivized elicitation, in each domain. The remaining four columns report the results from 10,000 simulation replications. In each replication, the sample is randomly divided into two groups. For each group we identify the qualitative self-assessment with the highest absolute correlation with the incentivized elicitation within the relevant domain. "Δ Correlation" is the difference between the magnitude of the estimated correlation in the subsample used to select the question and the estimated correlation in the other subsample, reported both in levels and as a percentage of the best correlation in full sample. The final two columns report the stability of the ranking between the two subsamples. "% Top" is the percentage of replications in which the same question was ranked first—and so would be selected—in both subsamples. The final column reports the Kendall's τ rank correlation, averaged across replications.

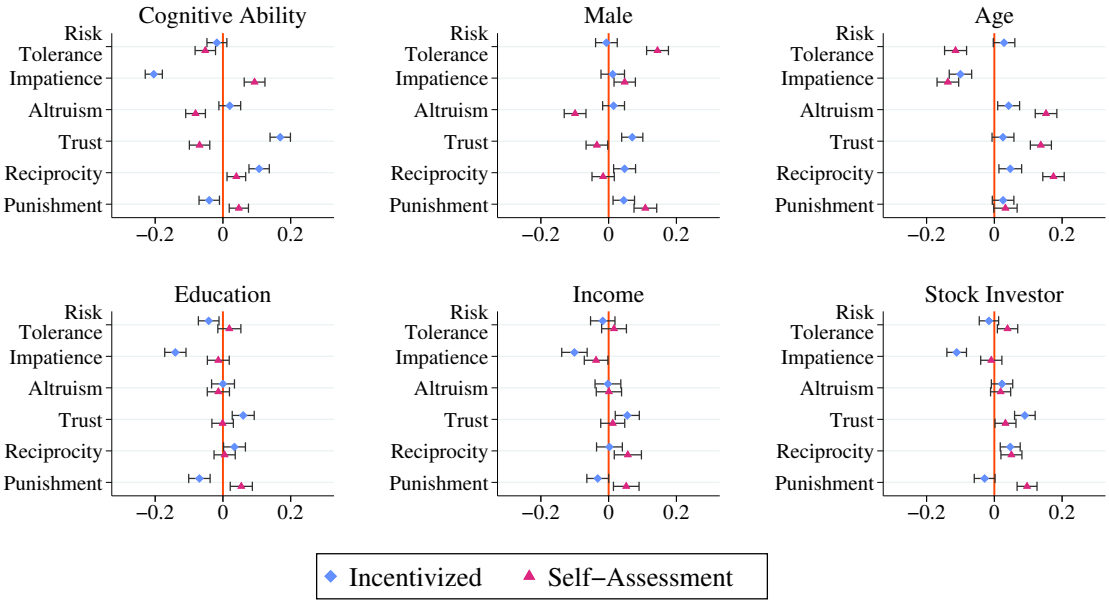
A.4 Correlations with Demographics

Figure A.3 displays univariate correlations between preferences and various individual characteristics.

Tables A.8 and A.9 display results for the same regressions as in Table 6 for the other four preference domains. In each case, the estimated relationships between qualitative self-assessments and demographic factors are essentially unchanged after controlling for incentivized elicitations. The only exception is the coefficient relating to cognitive ability and reciprocity, which becomes smaller and statistically insignificant.

Table A.10 shows results similar to those in Table 6 for risk tolerance and impatience using our two combined U.S. samples. For these two domains, we have duplicate elicitations in both samples. The results are similar to those reported in the main text. The only noteworthy

Figure A.3: Incentivized elicitations and qualitative self-assessments exhibit different correlations with individual characteristics.



Notes: Data from combined U.S. Samples 1 and 2 ($N = 4,950$). Each panel displays univariate correlations between a demographic characteristic and each preference measure. Bars represent 90% confidence intervals.

difference is that the correlation between qualitative risk tolerance and stock investment is now statistically significant, as documented by Dohmen et al. (2011). However, as with the other demographic factors, this relationship is not driven by preferences captured by the corresponding incentivized elicitation.

Figure A.4 compares the average preferences within our low-education U.K. Sample to those across the whole U.S. population.

Table A.8: Relationships between self-assessed impatience and trust and demographics are unrelated to variation in incentivized elicitations (N = 1,950).

	Dependent Variable = Qualitative Self-Assessment					
	Impatience			Trust		
Cognitive Ability	0.08***	0.08***	0.06**	−0.04	−0.08***	−0.06*
	(.029)	(.029)	(.030)	(.028)	(.027)	(.029)
Male	0.06	0.06	0.05	−0.05	−0.07	−0.06
	(.056)	(.056)	(.056)	(.056)	(.056)	(.056)
Age	−0.14***	−0.15***	−0.14***	0.16***	0.16***	0.17***
	(.029)	(.029)	(.029)	(.029)	(.028)	(.028)
Education	−0.05	−0.05	−0.05	−0.01	−0.01	−0.00
	(.031)	(.031)	(.031)	(.033)	(.032)	(.031)
Income (Log)	−0.05	−0.05	−0.07**	0.01	0.00	0.01
	(.035)	(.035)	(.034)	(.033)	(.034)	(.035)
Stock Investor	0.06	0.06	0.05	0.10	0.09	0.10
	(.069)	(.068)	(.067)	(.067)	(.066)	(.066)
Incentivized Elicitation:						
Risk Tolerance			−0.03			0.05
			(.037)			(.036)
Impatience		−0.01	−0.00			0.07*
		(.036)	(.036)			(.036)
Altruism			−0.05			0.21*
			(.116)			(.120)
Trust			0.26**		0.21***	0.11
			(.132)		(.047)	(.133)
Reciprocity			−0.14**			−0.09
			(.056)			(.054)
Punishment			0.01			−0.05
			(.039)			(.038)

Notes: All columns use data from U.S. Sample 1. Incentivized elicitations are instrumented to eliminate the effect of classical measurement error. Coefficients and standard errors (in parentheses) on all continuous measures are standardized. All specifications include an indicator variable for missing income. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

A.5 Correlations with Self-Reported Behaviors

In this section we examine correlations between qualitative self-assessments and self-reported behaviors, for a subset of U.S. Sample 2. Data for part of this sample was collected as part of

Table A.9: Relationships between demographics and self-assessed reciprocity and willingness-to-punish are unrelated to variation in incentivized elicitations (N = 1,950).

	Dependent Variable = Qualitative Self-Assessment					
	Reciprocity			Punishment		
Cognitive Ability	0.09***	0.06**	0.04	−0.01	−0.00	−0.00
	(.029)	(.028)	(.031)	(.027)	(.027)	(.029)
Male	−0.02	−0.04	−0.05	0.26***	0.25***	0.25***
	(.057)	(.057)	(.056)	(.059)	(.059)	(.059)
Age	0.20***	0.18***	0.18***	−0.02	−0.03	−0.03
	(.029)	(.029)	(.028)	(.031)	(.030)	(.030)
Education	−0.05	−0.05	−0.04	0.02	0.03	0.03
	(.030)	(.029)	(.029)	(.032)	(.032)	(.032)
Income (Log)	0.03	0.04	0.02	−0.02	−0.03	−0.03
	(.035)	(.033)	(.034)	(.036)	(.035)	(.036)
Stock Investor	0.08	0.08	0.06	0.12*	0.12*	0.12*
	(.063)	(.060)	(.058)	(.067)	(.069)	(.069)
Incentivized Elicitation:						
Risk Tolerance			−0.04			0.03
			(.039)			(.041)
Impatience			−0.06			0.00
			(.037)			(.037)
Altruism			0.24**			0.06
			(.111)			(.125)
Trust			0.07			0.02
			(.124)			(.143)
Reciprocity		0.21***	0.04			−0.04
		(.046)	(.053)			(.058)
Punishment			0.05		0.15***	0.14***
			(.036)		(.042)	(.041)

Notes: All columns use data from U.S. Sample 1. Incentivized elicitations are instrumented to eliminate the effect of classical measurement error. Coefficients and standard errors (in parentheses) on all continuous measures are standardized. All specifications include an indicator variable for missing income. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

a five-wave survey. The main text analyzes the first wave. In waves 3 to 5 of the survey, we elicited self-reported behaviors relating to charitable giving, community engagement, health

Table A.10: Findings regarding the relationships between qualitative self-assessments and demographics are similar using combined U.S. Samples ($N = 4{,}950$).

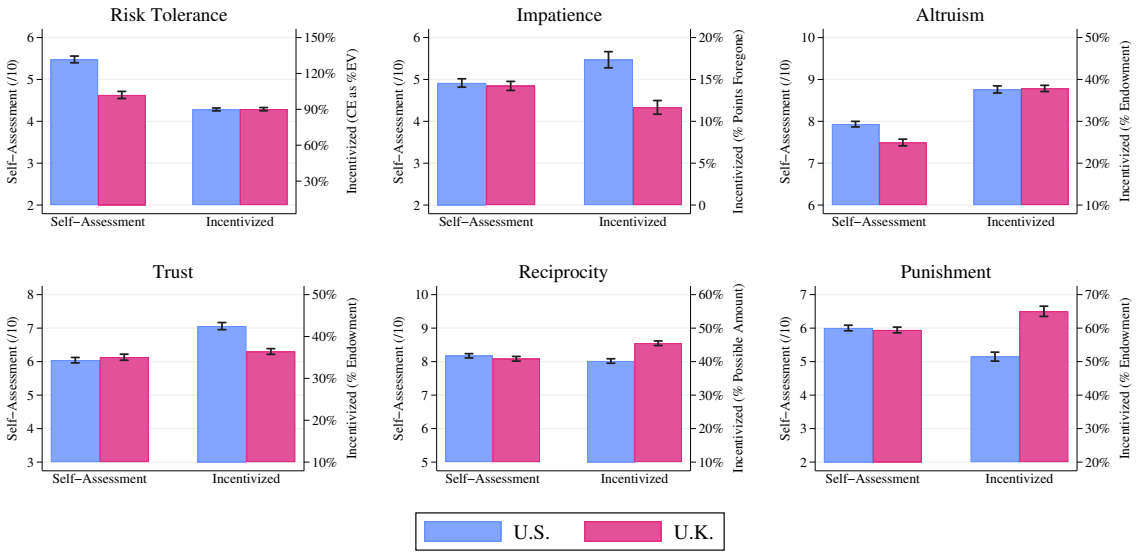
	Dependent Variable = Qualitative Self-Assessment			
	Risk Tolerance		Impatience	
Incentivized Measure		0.13^{***}		-0.01
		(.026)		(.026)
Cognitive Ability	-0.12^{***}	-0.12^{***}	0.09^{***}	0.09^{***}
	(.019)	(.019)	(.019)	(.020)
Male	0.31^{***}	0.31^{***}	0.05	0.05
	(.039)	(.039)	(.039)	(.039)
Age	-0.13^{***}	-0.14^{***}	-0.13^{***}	-0.13^{***}
	(.021)	(.021)	(.020)	(.020) ’
Education	0.04^{**}	0.05^{**}	-0.03	-0.03
	(.021)	(.021)	(.021)	(.021)
Income (Log)	0.01	0.01	-0.04^{*}	-0.04^{*}
	(.024)	(.024)	(.023)	(.023)
Stock Investor	0.12^{***}	0.13^{***}	0.05	0.05
	(.043)	(.043)	(.044)	(.044)

Notes: All columns use data from U.S. Samples 1 and 2. Incentivized elicitations are instrumented to eliminate the effect of classical measurement error. Coefficients and standard errors (in parentheses) on all continuous measures are standardized. All specifications include an indicator variable for missing income. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

behaviors, and political interest.[1] Some previous studies have suggested that self-assessments may be valuable based on their ability to predict such activities. Here, we examine the relationships between these behaviors, qualitative self-assessments, and incentivized preference measures in this survey.

[1]YouGov provides separate probability weights for each survey wave. We use sample weights from the first survey in waves 3 to 5 that a participant completed.

Figure A.4: Comparison of low-education U.K. Sample to U.S. general population.



Notes: The figure compares the average preference within the two U.S. representative samples ($N = 4,950$) and the U.K. Sample ($N = 1,984$).

Table A.11: Relationships between qualitative self-assessments and behaviors.

	Risk	Impatience	Altruism	Trust	Reciprocity	Punishment
Smoke	0.30***	0.08	0.16**	−0.04	0.21***	0.07
	(.078)	(.091)	(.079)	(.102)	(.074)	(.076)
Binge Drink	0.43***	0.20**	−0.02	0.18**	0.08	0.17**
	(.091)	(.082)	(.085)	(.079)	(.069)	(.078)
Eat 5 Fruit/Veg A Day	0.32***	−0.12*	0.07	0.11*	0.01	0.01
	(.064)	(.063)	(.064)	(.063)	(.065)	(.060)
Exercise	0.38***	−0.19***	0.15**	0.13**	0.13**	0.08
	(.063)	(.062)	(.066)	(.060)	(.065)	(.059)
Attended Local Political Meeting	0.32***	0.18**	0.27***	0.18**	0.22***	0.27***
	(.093)	(.078)	(.077)	(.085)	(.072)	(.078)
Put Up Political Sign	0.19**	0.15**	0.26***	0.08	0.30***	0.23***
	(.079)	(.077)	(.068)	(.082)	(.064)	(.073)
Worked for Political Campaign	0.33***	0.30***	0.28***	0.17	0.13	0.18*
	(.110)	(.111)	(.098)	(.119)	(.102)	(.094)
Donated to Campaign	0.30***	0.18***	0.29***	0.14**	0.33***	0.24***
	(.068)	(.065)	(.066)	(.068)	(.052)	(.058)
Donated Blood	0.40***	0.08	0.30***	0.20**	0.26***	0.10
	(.097)	(.101)	(.086)	(.096)	(.089)	(.095)
Worked for Community	0.36***	0.17***	0.41***	0.09	0.26***	0.18***
	(.069)	(.065)	(.057)	(.071)	(.060)	(.067)
Contacted Govt Official	0.19***	0.23***	0.34***	0.03	0.26***	0.22***
	(.063)	(.063)	(.056)	(.069)	(.056)	(.051)
Community Meeting	0.45***	0.19***	0.37***	0.19**	0.28***	0.19***
	(.074)	(.068)	(.058)	(.074)	(.058)	(.070)
Volunteer Work	0.00	0.00	0.01***	0.00	0.01***	0.00
	(.002)	(.002)	(.003)	(.002)	(.003)	(.002)
Church/Charity Contribution	0.23***	−0.08	0.59***	0.31***	0.44***	−0.03
	(.066)	(.061)	(.061)	(.058)	(.062)	(.058)
Church Attendance	0.12***	−0.00	0.25***	0.18***	0.12***	−0.01
	(.033)	(.033)	(.030)	(.031)	(.028)	(.030)
# Organizations Member of	0.17***	0.08***	0.15***	0.09***	0.11***	0.09***
	(.026)	(.025)	(.029)	(.026)	(.029)	(.025)
#Days Talked Politics	0.07**	0.02	0.20***	0.09***	0.23***	0.06**
	(.030)	(.029)	(.034)	(.030)	(.027)	(.026)
Read Blog	0.14**	0.22***	0.13**	0.05	0.20***	0.08
	(.063)	(.061)	(.061)	(.063)	(.059)	(.059)
Watched TV News	0.08	−0.11*	0.35***	0.31***	0.31***	0.11*
	(.065)	(.062)	(.072)	(.065)	(.070)	(.063)
Read Newspaper	0.26***	−0.03	0.21***	0.24***	0.28***	0.02
	(.060)	(.060)	(.058)	(.058)	(.057)	(.053)
Listened to Radio	0.19***	0.12**	0.23***	0.14**	0.13**	0.12**
	(.064)	(.057)	(.059)	(.054)	(.064)	(.053)

Notes: All columns use data from U.S. Sample 2, waves 3–5 ($N = 4,134$). Coefficients and standard errors (in parentheses) on all continuous measures are standardized. Standard errors are clustered by participant. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

First, Table A.11 presents simple correlations between each of the qualitative self-assessments and different self-reported activities. Most variables in this table are binary, taking a value of one if a participant reported an activity, and zero otherwise. Exceptions include church attendance, the (square-root of the) number of organizations a participant is a member of, and the number of days a participant spent discussing politics, where these variables are all standardized. Since some individuals responded to multiple waves of the survey, we cluster standard errors by participant.

As can be seen, qualitative self-assessments are correlated with a wide range of behaviors, sometimes in unexpected ways. For instance, we replicate the finding of Dohmen et al. (2011) that self-reported willingness to take risks is correlated with smoking.[2] However, it is also correlated with almost every other behavior in our dataset, including the seemingly low risk behavior of eating five portions of fruit and vegetables per day. We also see that the social preference measures are correlated with a wide range of activities relating to charitable activity, engagement with local politics, and political interest.

To aid interpretation of this large battery of variables, we carry out two principal components analyses—see Figure A.5 and Tables A.12–A.13. In each analysis, we determine the number of components to retain using parallel analysis. The first principal components analysis includes the four variables, at the top of Table A.11, related to health behaviors. This analysis, reported in Table A.12, identifies two intuitive components—*Unhealthy Behaviors* (primarily capturing smoking and drinking more than five alcoholic drinks) and *Healthy Behaviors* (loading on exercise, eating five fruit and/or vegetables a day). The second analysis, reported in Table A.13, includes the remaining variables, which relate to involvement in a participant's local community, volunteering or giving to charity, political activity, and media use. This identifies three components. *Political Engagement* primarily relates to political activities such as working for a political candidate or working to tackle issues in the local community. *Charity/Community* relates primarily to volunteering, religious attendance, and

[2]See Arslan et al. (2020, Supplementary Materials S1) for a general review of studies assessing relationships between the qualitative self-assessment of risk and behaviors.

Figure A.5: Scree Plots for Principal Components Analyses.

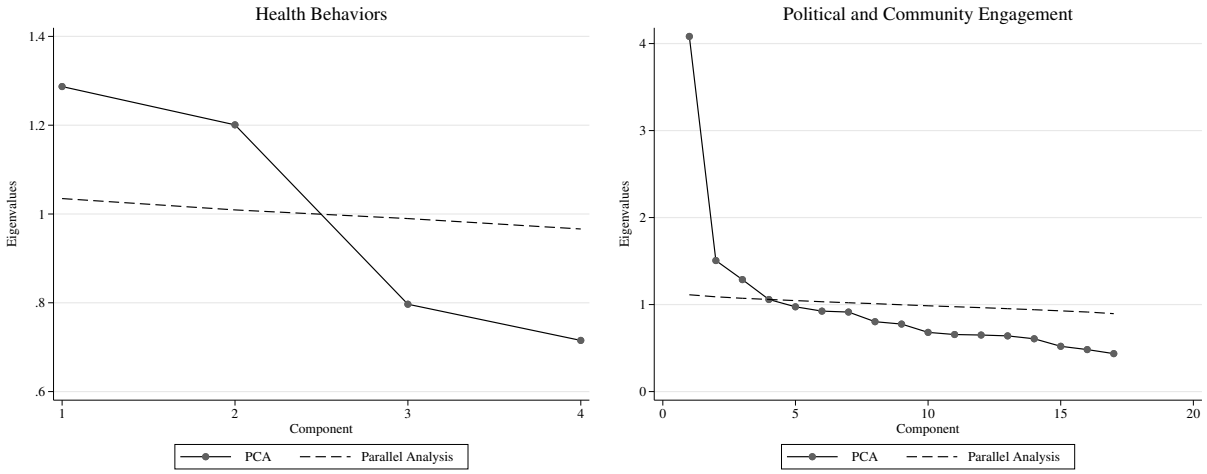


Table A.12: Principal Components Analysis for Health Behaviors.

	Unhealthy Behaviors	*Healthy Behaviors*	Unexplained
Do you currently...			
Smoke cigarettes, cigars, or pipes	0.72	−0.12	0.34
Drink five or more alcoholic drinks occasionally	0.70	0.13	0.36
Exercise	0.00	0.69	0.42%
Eat five or more servings of fruits and/or vegetables a day	−0.01	0.70	0.40
Percent of Variation	32%	31%	38%

Notes: The principal components analysis used data from waves 3–5 of U.S. Sample 2 ($N = 4,134$). Each question offered participants a choice of yes or no.

contributing to church and charity. *Political Interest* captures variance relating to the use of media, and discussing politics with friends and family.

Figure A.6 shows correlations between these five principal components and both incentivized elicitations and qualitative self-assessments. Both types of measures are correlated with several behavioral variables, but the patterns are markedly different, particularly for

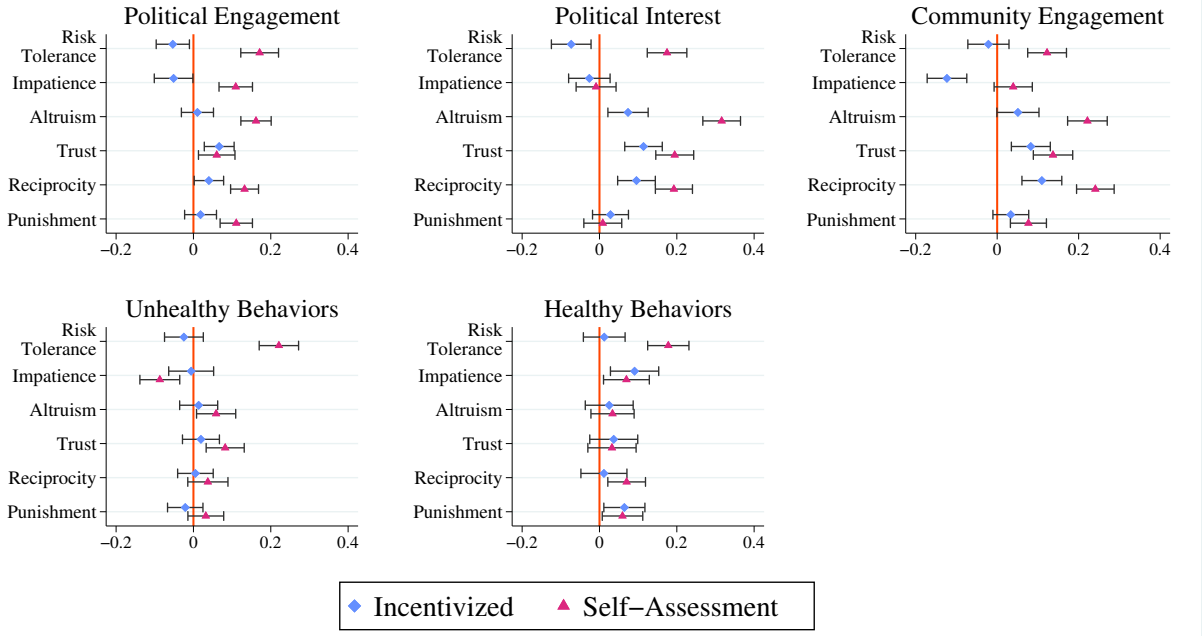
Table A.13: Principal Components Analysis for Political and Community Activities.

	Political Engagement	*Charity/ Community*	*Political Interest*	Unexplained
During the past year did you...				
Attend local political meetings	0.43	−0.03	−0.05	0.48
Put up a political sign	0.33	0.07	−0.16	0.65
Work for a candidate or campaign	0.40	−0.07	−0.08	0.57
Donate money to a candidate, campaign, or political organization	0.32	0.15	−0.14	0.60
Donate blood	0.14	−0.12	0.21	0.85
Indicate whether in past 12 months you...				
Worked with other people to deal with some issue facing your community	0.35	−0.01	0.14	0.52
Telephoned, wrote a letter to, or visited a government official to express your views on a public issue	0.28	0.16	−0.01	0.61
Attended a meeting about an issue facing your community or schools	0.36	−0.05	0.12	0.54
In past 12 months devoted time or made contributions...				
Did volunteer work	−0.04	0.14	0.08	0.95
Contributed to church and charity	0.01	0.10	0.56	0.36
How often attend church	−0.04	−0.05	0.65	0.33
# of organizations a member of	0.25	−0.02	0.31	0.55
In past 24 hours have you.....				
Read a blog	−0.03	0.34	0.13	0.71
Watched TV news	0.05	0.40	−0.12	0.66
Read a newspaper in print or online	−0.00	0.47	0.01	0.55
Listened to a radio news program or talk radio	−0.16	0.41	0.08	0.69
# Days in last week discussed politics with friends/family	0.06	0.47	−0.01	0.50
Percent of Variation	18%	12%	10%	60%

Notes: The principal components analysis used data from waves 3–5 of U.S. Sample 2 ($N = 4,134$).

risk and impatience. Interestingly, however, both incentivized and self-assessment measures of altruism, trust, and reciprocity have predictive power for behaviors related to political engagement, political interest, and charitable giving/community engagement. Bigger dif-

Figure A.6: Correlations between principal components of self-reported behaviors and preferences.



Notes: The figure uses data from U.S. Sample 2, waves 3–5 ($N = 4,134$). Standard errors are clustered by participant. Bars represent 90% confidence intervals.

ferences appear in the realm of risk and, to a lesser extent, impatience. This indicates that incentivized measures may have better predictive power than might be suggested by examining risk measures alone (Charness et al., 2020).

Finally, in Table A.14, we present regression results similar to those in Table 6. That table indicated whether correlations between demographic characteristics and qualitative self-assessments capture behavior on incentivized elicitations. Here, we investigate whether the correlations with demographics are explained by a participant's behavior outside of the survey—if so, this could suggest that self-assessments are capturing some other latent factor that predicts behavior, but is not captured in incentivized elicitations. We also test whether the correlations between qualitative self-assessments and self-reported behaviors are explained by the incentivized elicitation. The surveys that included questions about behaviors included only a single incentivized elicitation of altruism (as with the initial survey in U.S. Study 2 that we study in the main text). As such, to account for measurement error,

we use the incentivized elicitation of trust as an instrument for altruism. The first stage F-statistic is 96.2, suggesting that this is appropriate.[3]

The results in Table A.14 show that, consistent with the correlations in Table A.11, the risk and altruism measures are correlated with a range of behaviors, not only those most obviously related to each preference. Second, with the possible exception of education, the coefficients related to demographic variables are little affected by the inclusion of self-reported behaviors. This suggests that the correlations between demographics and self-assessments are not reflected in different behaviors outside the survey. For instance, if women appear more altruistic based on self-assessments, this does not translate into increased giving in the dictator game or higher (self-reported) charitable contributions. Third, the correlations between self-reported behaviors and self-assessments are close to unchanged after controlling for the incentivized measures. That is, the variation between self-reported behaviors and qualitative self-assessments is not correlated with variation in the incentivized elicitations.

[3]The incentivized risk tolerance elicitation included in these surveys was slightly different from the elicitation in the original survey of U.S. Sample 2, which we analyze in the main text.

Table A.14: Relationships between qualitative self-assessments, demographics, and behaviors.

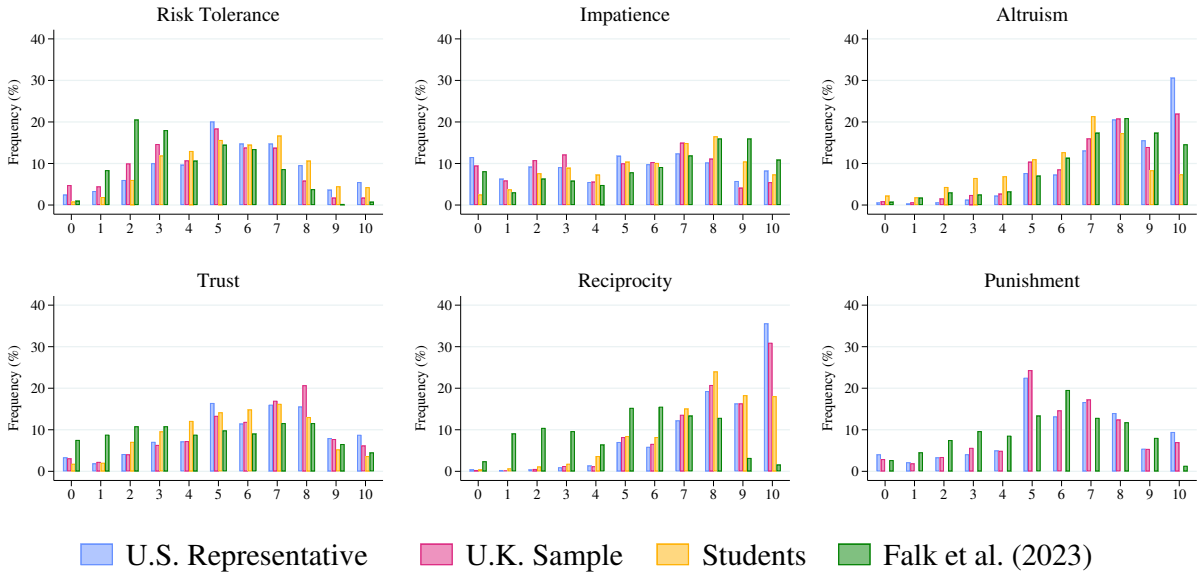
	Dependent Variable = Qualitative Self-Assessment					
	Risk Tolerance			Altruism		
Cognitive Ability	−0.12***	−0.10***	−0.09***	0.01	0.01	0.00
	(.031)	(.028)	(.028)	(.030)	(.029)	(.029)
Male	0.23***	0.18***	0.17***	−0.20***	−0.22***	−0.23***
	(.069)	(.063)	(.063)	(.067)	(.060)	(.059)
Age	−0.08**	−0.10***	−0.10***	0.13***	0.05	0.04
	(.033)	(.033)	(.033)	(.031)	(.031)	(.031)
Education	0.10***	0.04	0.04	−0.01	−0.09***	−0.08***
	(.035)	(.035)	(.035)	(.032)	(.031)	(.030)
Income (Log)	0.07*	0.03	0.03	0.11***	0.06*	0.07**
	(.037)	(.033)	(.033)	(.037)	(.034)	(.034)
Behaviors:						
Unhealthy Activities		0.15***	0.15***		−0.02	−0.02
		(.031)	(.031)		(.028)	(.028)
Healthy Activities		0.16***	0.16***		0.05*	0.05
		(.031)	(.031)		(.030)	(.030)
Political Engagement		0.08***	0.08***		0.05**	0.06**
		(.023)	(.023)		(.025)	(.025)
Political Interest		0.12***	0.12***		0.27***	0.26***
		(.029)	(.029)		(.030)	(.031)
Charity / Community		0.06*	0.05*		0.12***	0.11***
		(.031)	(.031)		(.035)	(.036)
Incentivized Measure			0.06*			0.13***
			(.033)			(.046)

Notes: All columns use data from U.S. Sample 2, waves 3–5 ($N = 4,134$). Incentivized measures are instrumented to eliminate the effect of classical measurement error. Coefficients and standard errors (in parentheses) on all continuous measures are standardized. All specifications include an indicator variable for missing income. ***, **, * denote statistical significance at the 1%, 5%, and 10% level.

A.6 Responses to Qualitative Self-Assessments

Figure A.7 shows the distribution of responses to each qualitative self-assessment across the different samples. Figure A.8 presents the percentage of responses at each focal value,

Figure A.7: Frequency of responses to qualitative self-assessments by sample.



Notes: Each panel relates to a different qualitative self-assessment, with bars reflecting the share of participants choosing each value in each sample. U.S. Representative includes U.S. Samples 1 and 2 (combined $N = 4,950$). Students combines Caltech, Pittsburgh, and UBC samples.

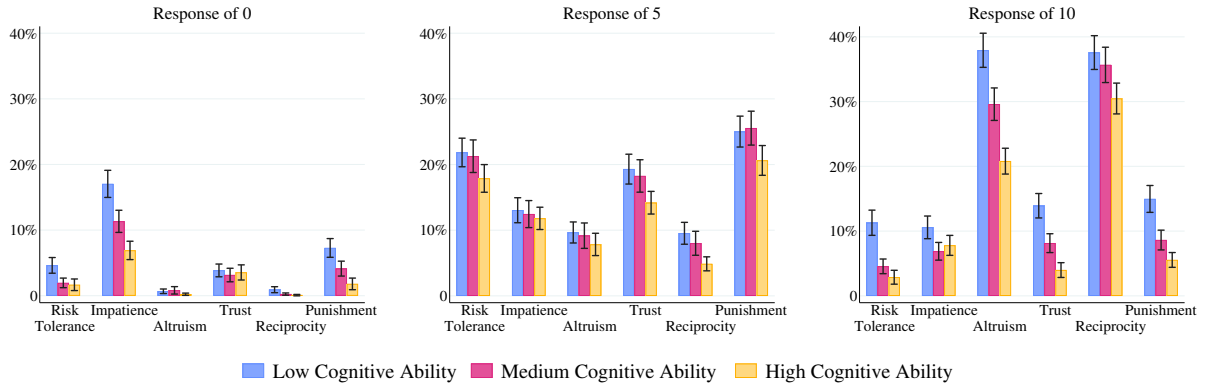
disaggregated by level of cognitive ability.

B Data Sources and Screenshots

B.1 General Population Datasets

Our general population datasets are drawn from a number of incentivized online surveys. Each survey was conducted by YouGov, a commercial survey company. Participants in the three representative samples were drawn from YouGov's two-million-person survey panel. YouGov obtains nationally representative samples by using targeted quota sampling and then constructing sample weights to account for the fact that some demographic groups are underrepresented. See Chapman et al. (2025b) for further discussion of the composition of the YouGov survey panel and payment procedure. Typically, each survey took around 45 minutes to one hour to complete, with payment approximately three times the average for similar surveys. In each survey, participants were paid for either one or two randomly-

Figure A.8: Response Shares at Each Focal Value, by Cognitive Ability



Notes: Each panel displays the percentage of responses that were at each focal value—0, 5, or 10—in each qualitative self-assessment. Low, medium, and high cognitive ability refer to terciles, from the combination of U.S. Samples 1 and 2. Bars represent 90% confidence intervals.

selected questions.

B.2 Data Sources and Screenshots

In this section, we provide illustrative screenshots from our various samples. These are broken into two groups. One contains the U.S. Samples, U.K. Sample, and Pitt student sample, which all used the same survey platform, and thus had a similar look and feel, as well as identical incentive levels. The Caltech, UBC, and MTurk samples were on a different survey platform, and those measures thus had a different look and feel.

In the U.S., U.K., and Pitt samples, rewards for incentivized questions were expressed via "points", an internal YouGov currency used to pay panel members. These points can be converted into monetary compensation or prizes, using the approximate rate of $0.001 per point. YouGov allows points to be converted to awards at specific point values, which leads to a slightly convex payoff schedule. This convexity does not appear to distort behavior—see Chapman et al. (2025b, Appendix C.6) for a detailed discussion. In the Caltech, UBC, and MTurk samples, rewards for incentivized questions were expressed via "tokens." At Caltech, each 100 tokens translated to $1, while at UBC, each 100 tokens translated to $1CAD. On MTurk, where payments are generally lower, 300 tokens translated to $1.

Figure B.1: Qualitative Self-Assessment of Risk Tolerance (with 8 selected, U.S. and U.K.)

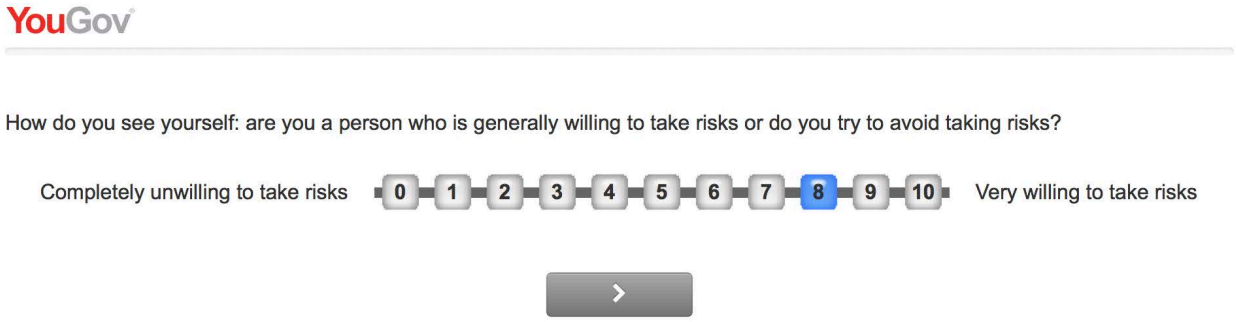


Figure B.2: Qualitative Self-Assessment of Impatience (U.S. and U.K.)

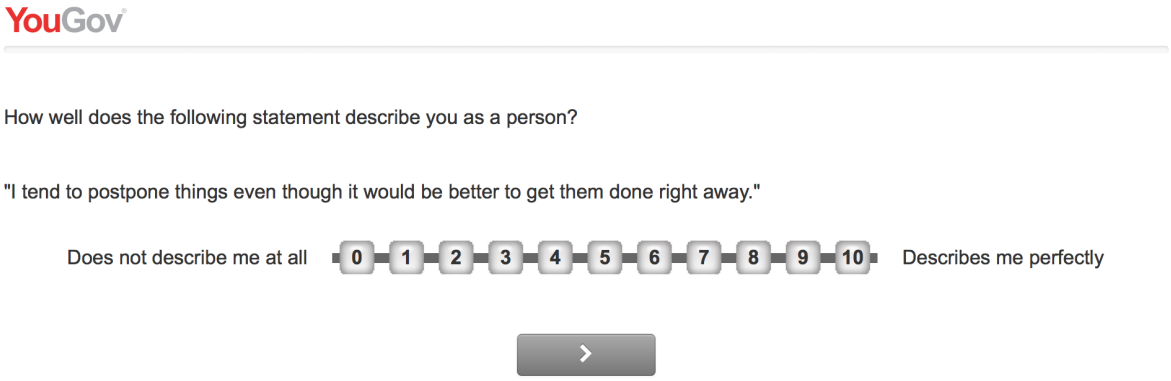


Figure B.3: Qualitative Self-Assessment of Altruism (U.S., U.K., and Pitt)

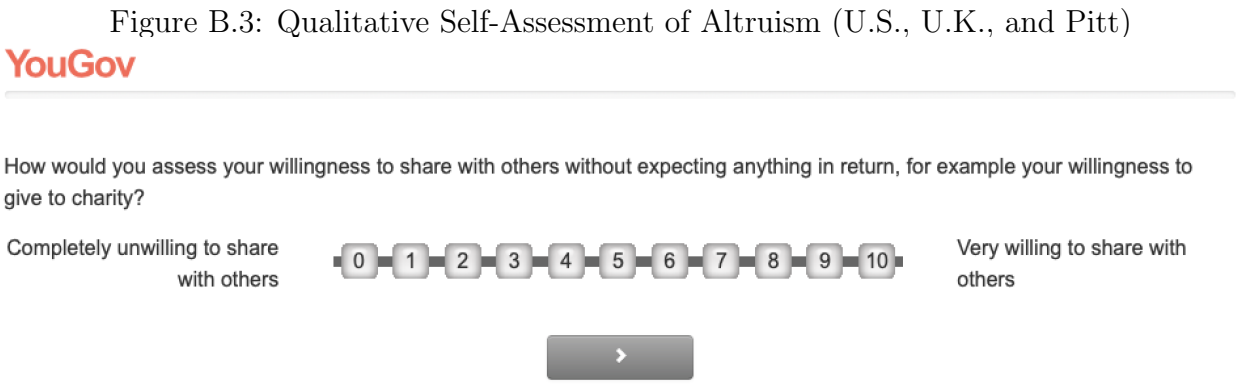


Figure B.4: Qualitative Self-Assessment of Trust (U.S., U.K., and Pitt)

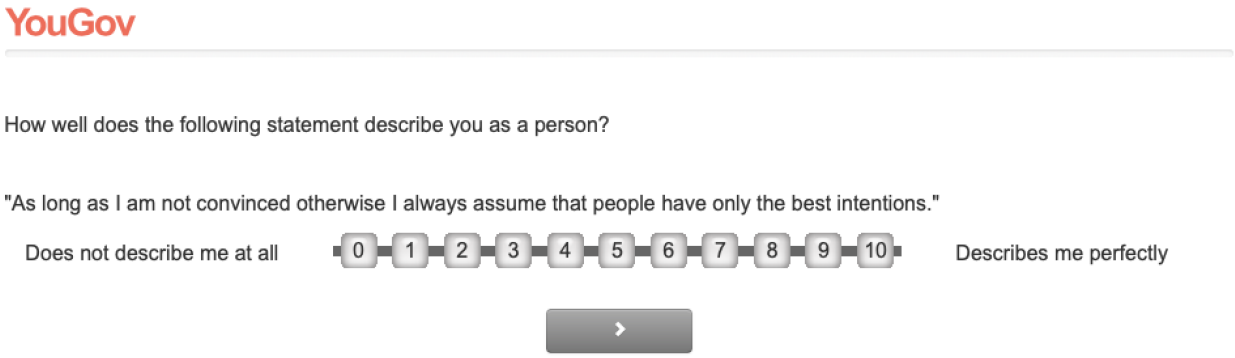


Figure B.5: Qualitative Self-Assessment of Reciprocity (U.S., U.K., and Pitt)

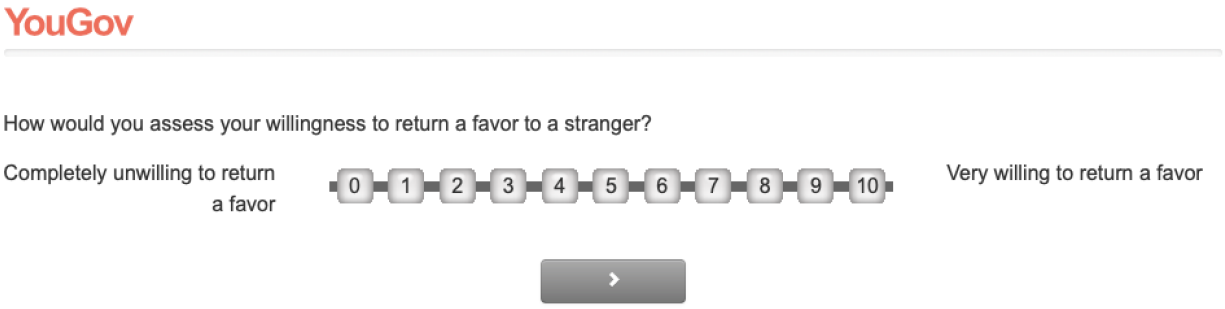


Figure B.6: Qualitative Self-Assessment of Punishment (U.S. and U.K.)

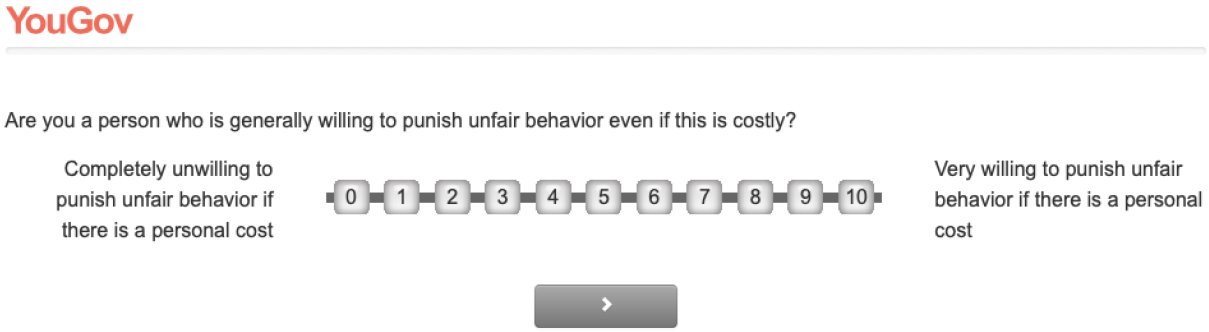


Figure B.7: Incentivized Elicitation of Risk Tolerance (U.S., U.K., and Pitt)

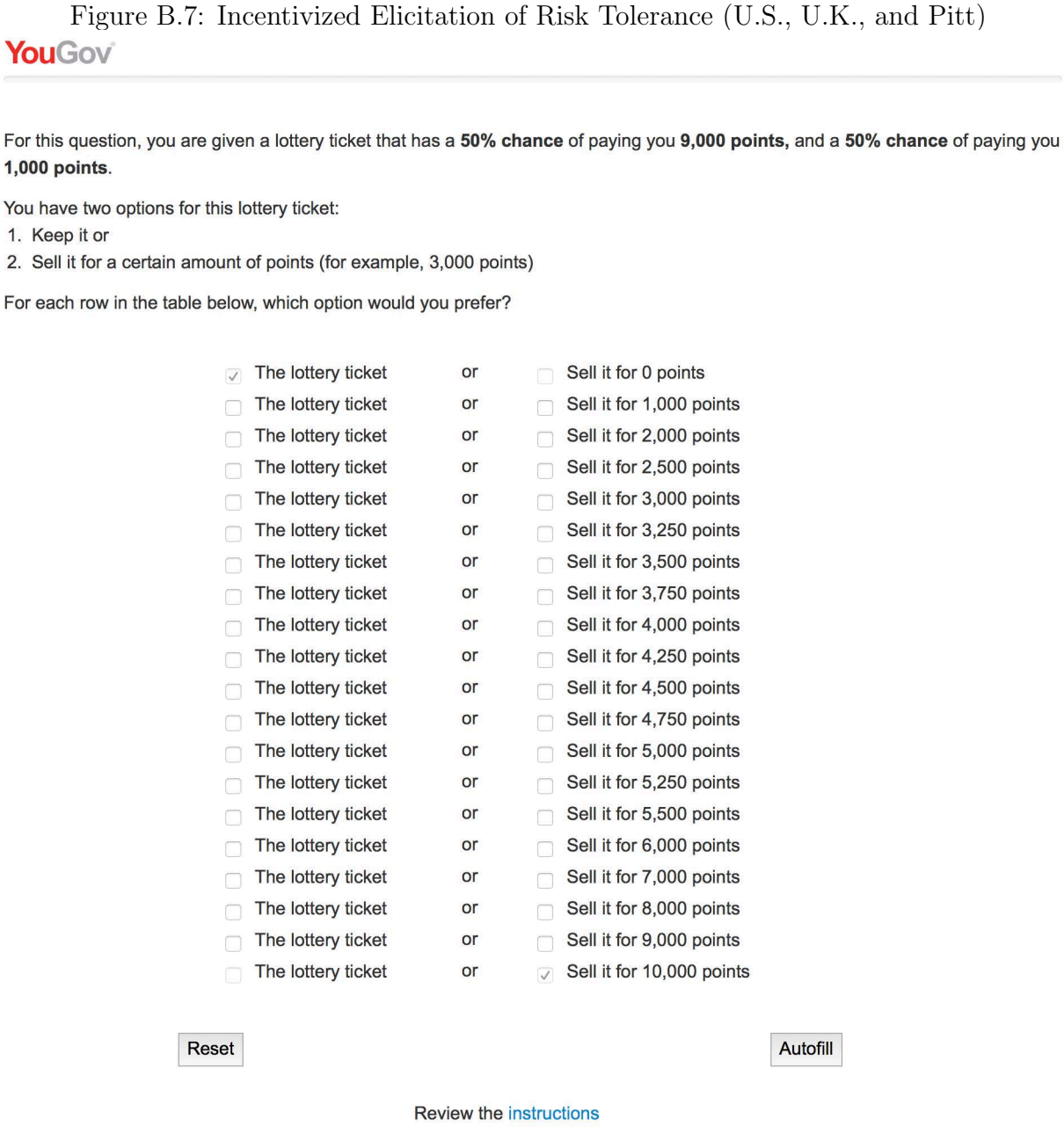


Figure B.8: Incentivized Elicitation of Impatience (U.S. and U.K.)

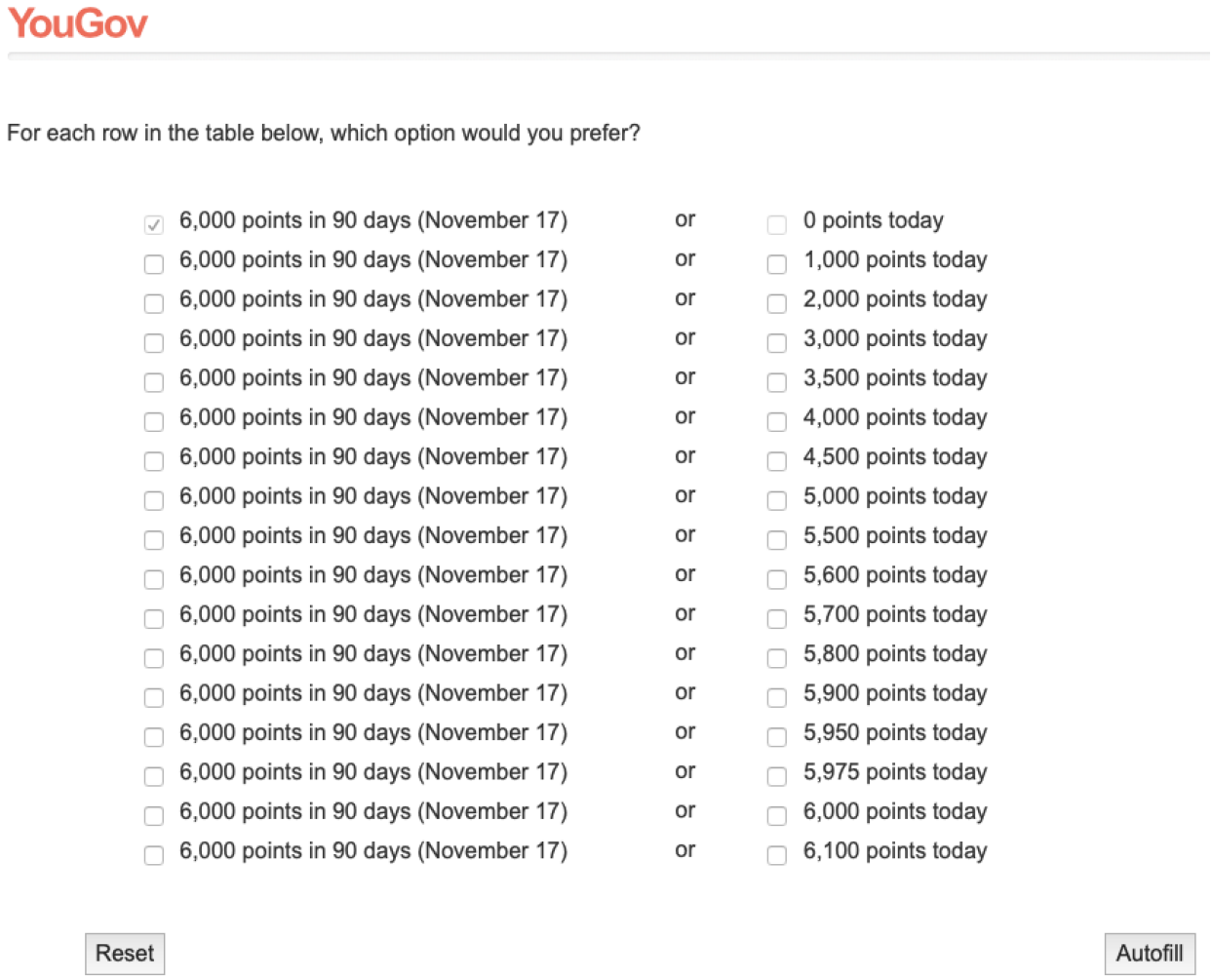


Figure B.9: Incentivized Elicitation of Altruism (U.S. and U.K.)

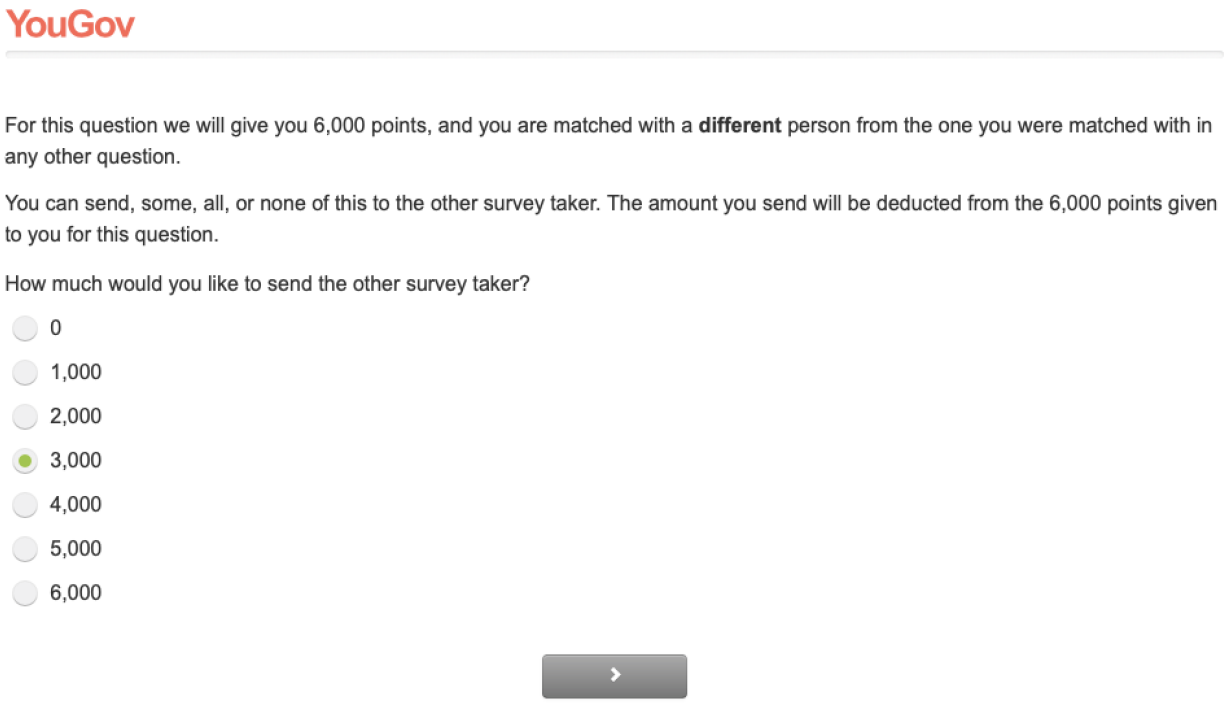


Figure B.10: Incentivized Elicitation of Trust (U.S. and U.K.)

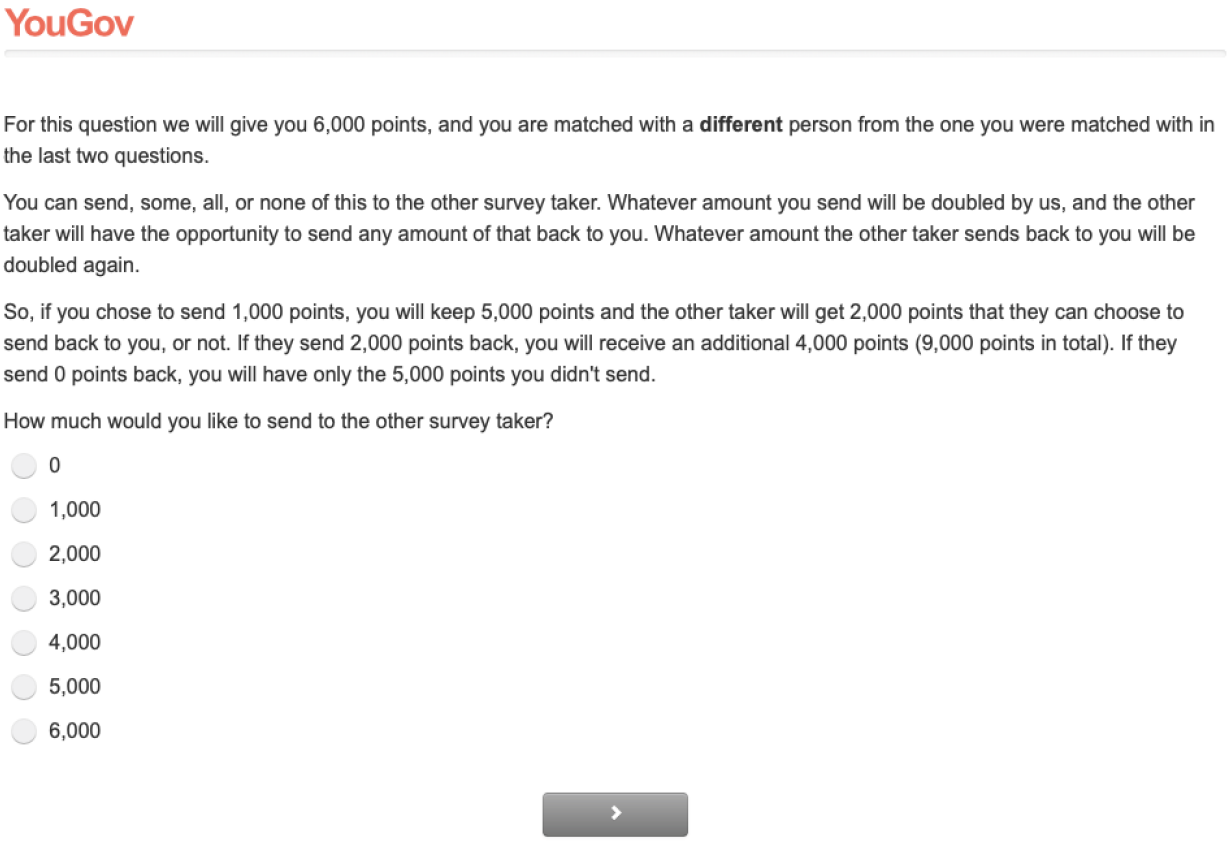


Figure B.11: Incentivized Elicitation of Reciprocity (U.S. and U.K.)

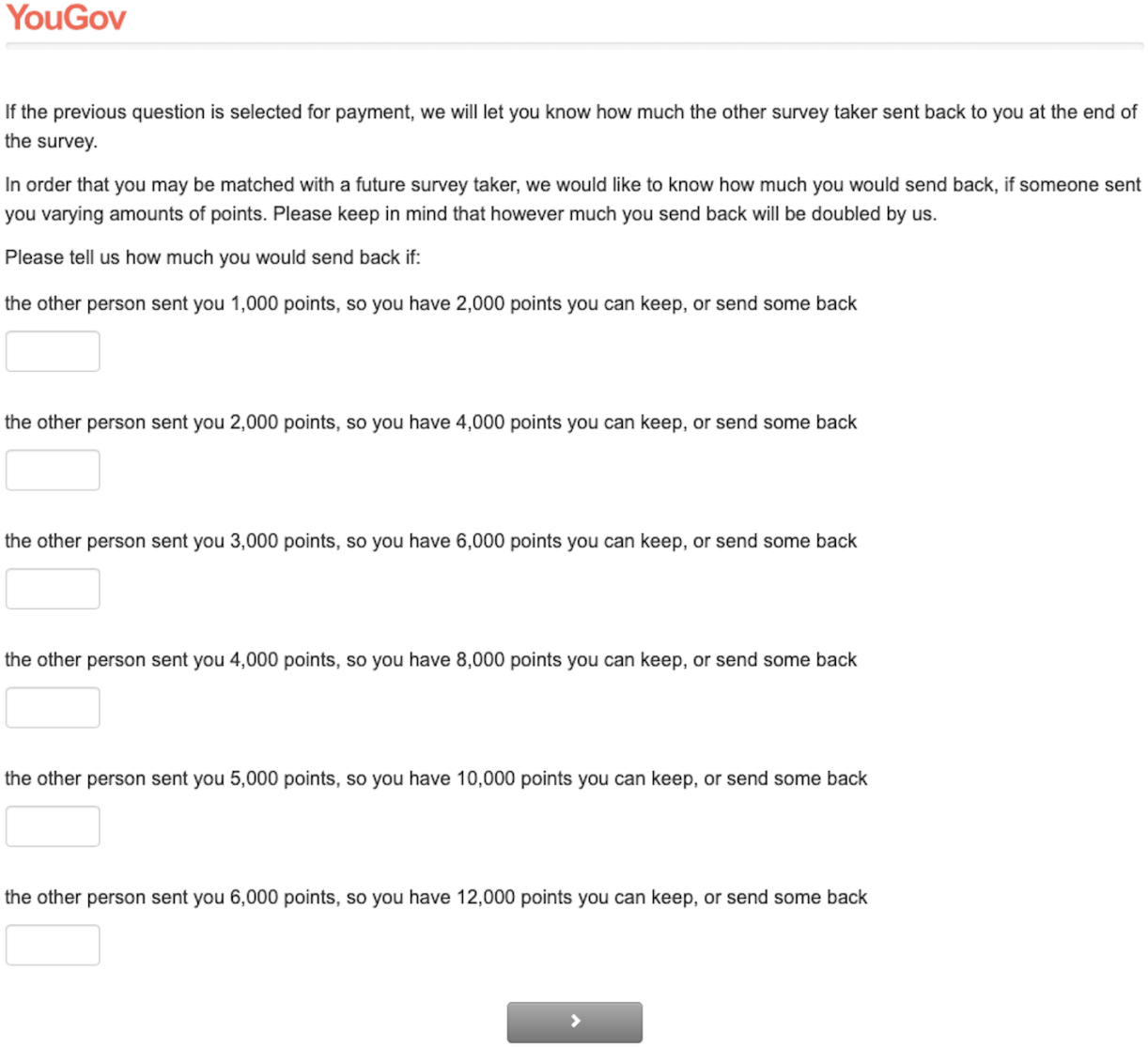


Figure B.12: Incentivized Elicitation of Punishment (U.S. and U.K.)

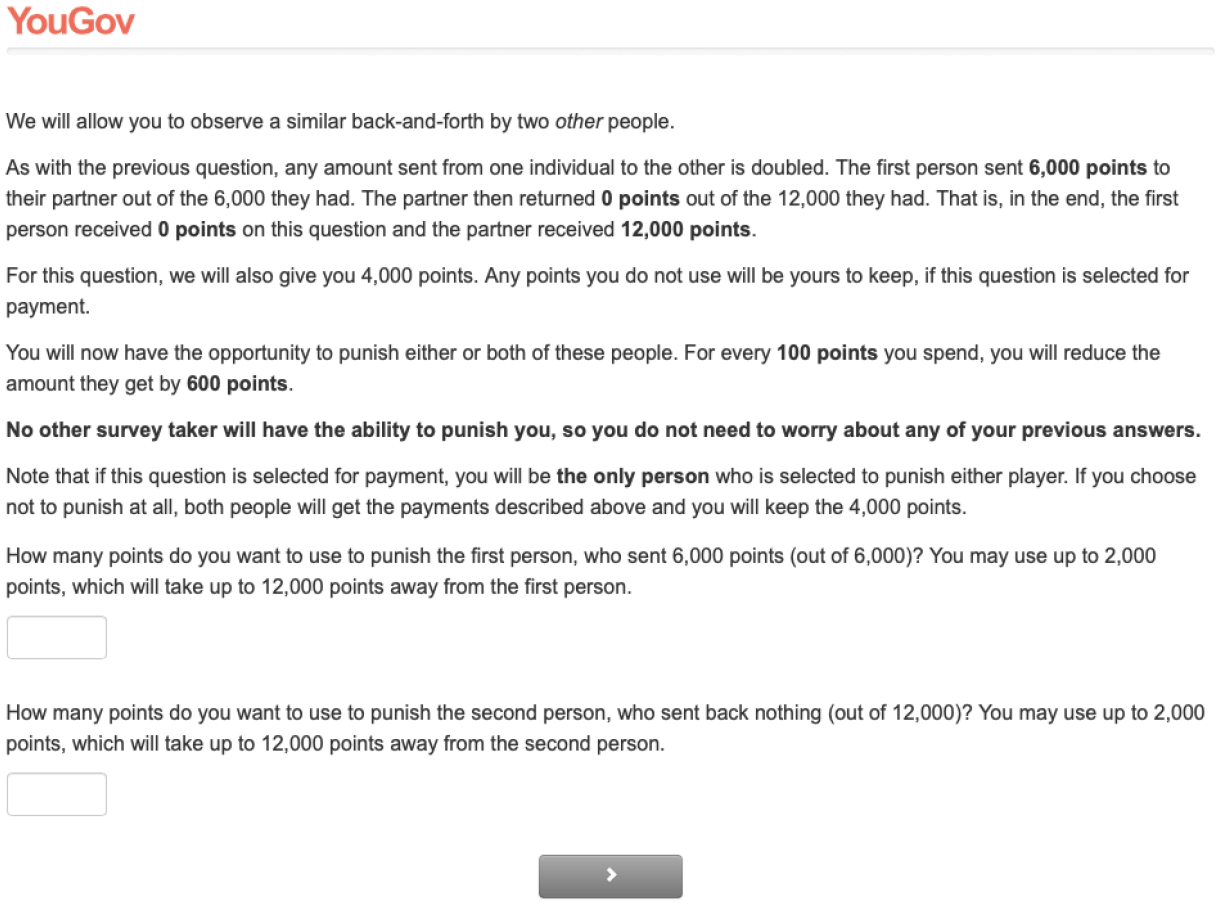


Figure B.13: Qualitative Self-Assessment of Risk Tolerance (Caltech, UBC, and MTurk)

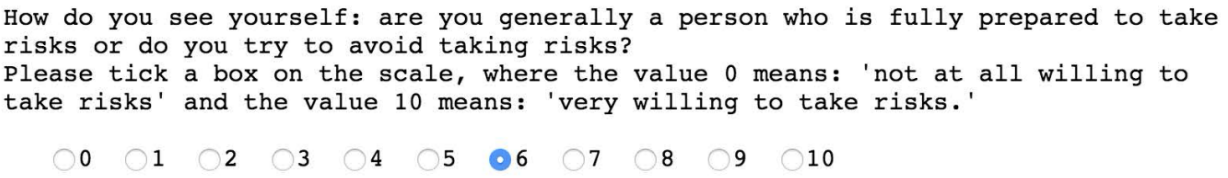


Figure B.14: Qualitative Self-Assessment of Impatience (Caltech)

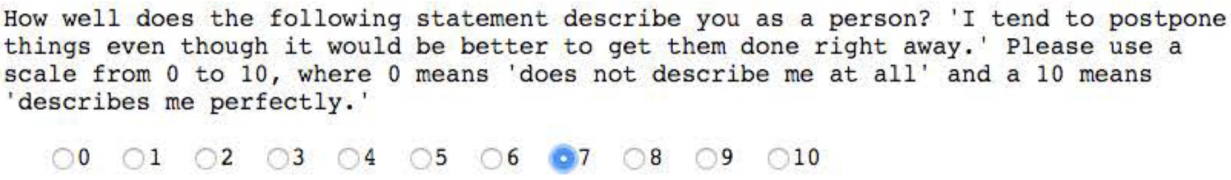


Figure B.15: Qualitative Self-Assessment of Altruism (Caltech, UBC, and MTurk)

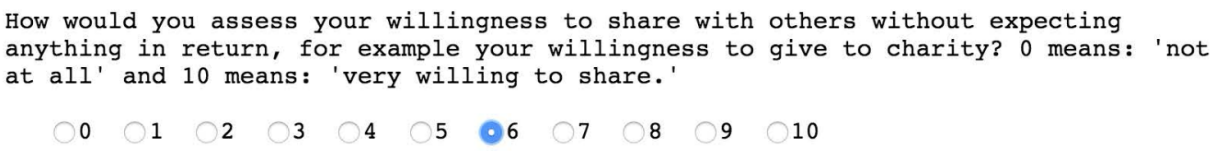
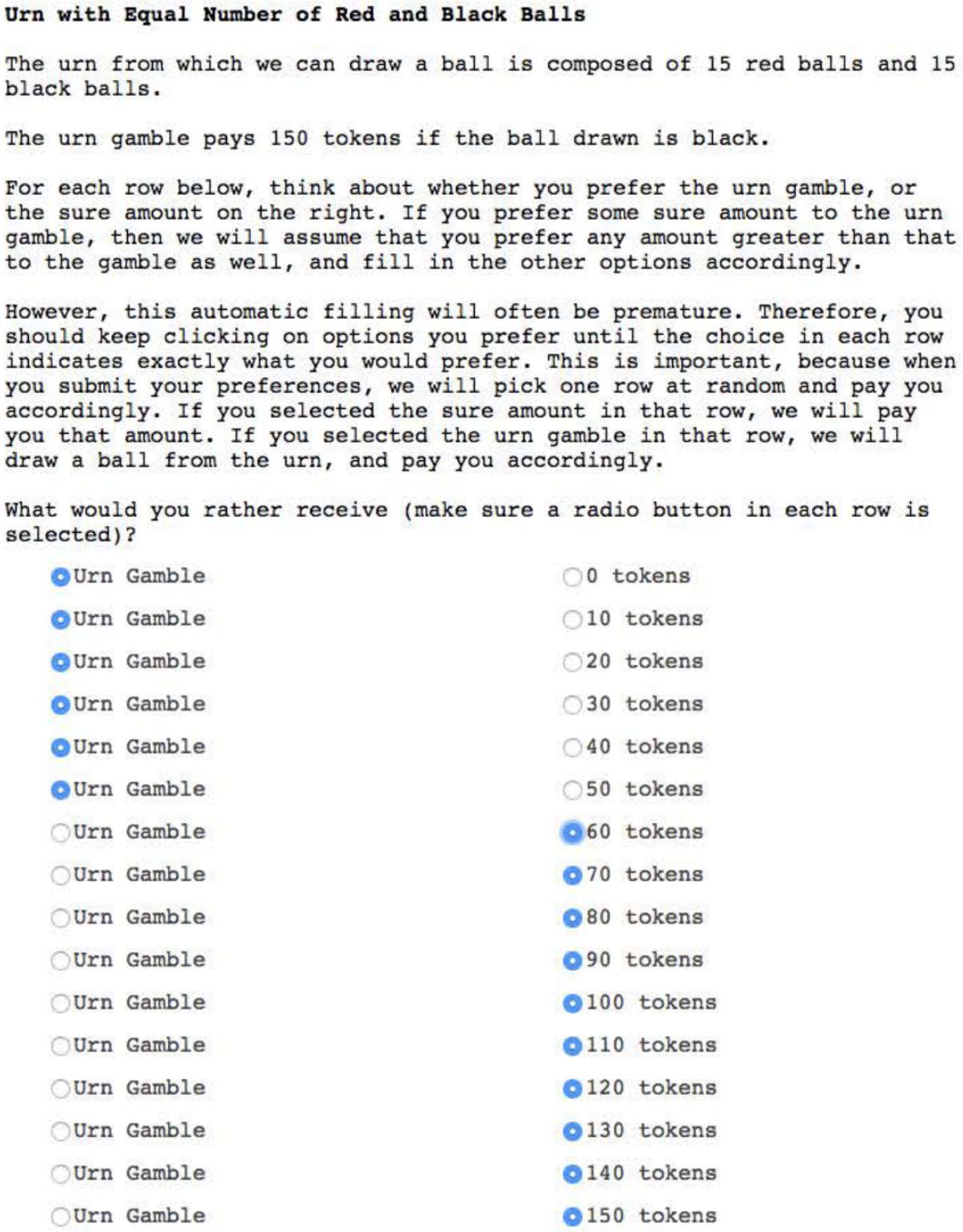


Figure B.16: Incentivized Elicitation of Risk Tolerance (Caltech, UBC, and MTurk)



Urn with Equal Number of Red and Black Balls

The urn from which we can draw a ball is composed of 15 red balls and 15 black balls.

The urn gamble pays 150 tokens if the ball drawn is black.

For each row below, think about whether you prefer the urn gamble, or the sure amount on the right. If you prefer some sure amount to the urn gamble, then we will assume that you prefer any amount greater than that to the gamble as well, and fill in the other options accordingly.

However, this automatic filling will often be premature. Therefore, you should keep clicking on options you prefer until the choice in each row indicates exactly what you would prefer. This is important, because when you submit your preferences, we will pick one row at random and pay you accordingly. If you selected the sure amount in that row, we will pay you that amount. If you selected the urn gamble in that row, we will draw a ball from the urn, and pay you accordingly.

What would you rather receive (make sure a radio button in each row is selected)?

● Urn Gamble	○ 0 tokens
● Urn Gamble	○ 10 tokens
● Urn Gamble	○ 20 tokens
● Urn Gamble	○ 30 tokens
● Urn Gamble	○ 40 tokens
● Urn Gamble	○ 50 tokens
○ Urn Gamble	● 60 tokens
○ Urn Gamble	● 70 tokens
○ Urn Gamble	● 80 tokens
○ Urn Gamble	● 90 tokens
○ Urn Gamble	● 100 tokens
○ Urn Gamble	● 110 tokens
○ Urn Gamble	● 120 tokens
○ Urn Gamble	● 130 tokens
○ Urn Gamble	● 140 tokens
○ Urn Gamble	● 150 tokens

Figure B.17: Incentivized Elicitation of Impatience (Caltech)

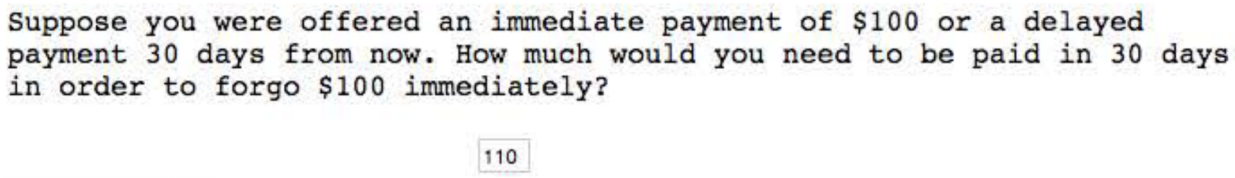


Figure B.18: Incentivized Elicitation of Altruism (Caltech, UBC, and MTurk)

